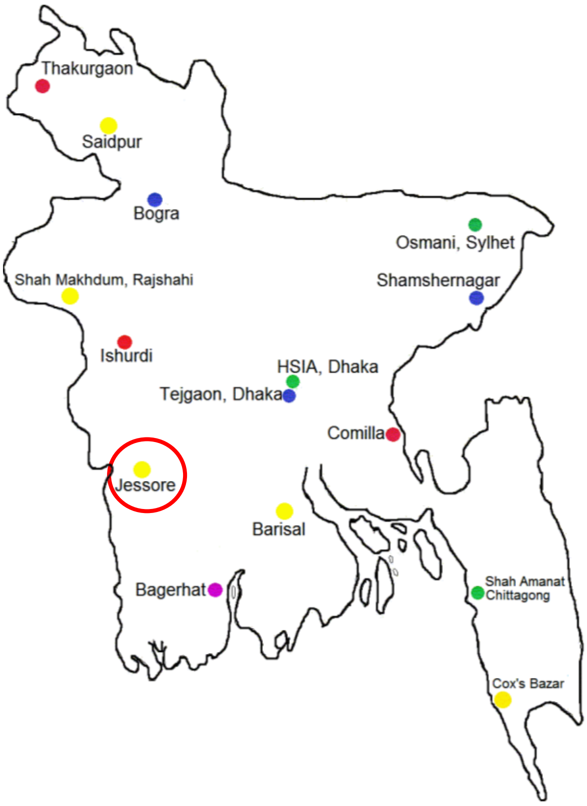
BAF BASE BIR-SRESHTO MATIUR RAHMAN



"CLIMATE OF JESSORE"

By

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November 2014

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PREFACE

1. BAF Base Bir Sreshto Matiur Rahman (BAF MTR) generates highest flying hours of BAF compare to other bases; which are mostly concern with training of cadets and U/T officers. Moreover, a number of occasions in the base, where very high officials including head of the state graces, make the Base different than any other bases of BAF. Almost all of these occasions are highly dependent upon weather conditions. Everyone agrees that AOR (Area of Responsibility) of BAF MTR experiences extreme weather than any other Bases. Therefore, requirement of reliable Wx forecast for this Base is everybody's concern in BAF. A weather forecast is simply a scientific estimate of future weather condition. Weather condition is the state of the atmosphere at a given time expressed in terms of the most significant weather variables. The significant weather variables being forecast differ from place to place and time to time. Therefore, in forecasting the weather, a Meteorologist must at least know something about the existing weather condition over a large area before he can make a reliable forecast. The accuracy of his forecast of a particular area depends largely upon his knowledge of the prevailing weather conditions and the climatology of that area. BAF MTR is acting as a Main Meteorological Office (MMO) of Bangladesh Meteorological Department (BMD) and it has a well heeled data bank. After getting the opportunity to work in BAF MTR, the desire of utilization of these data was attempted and the "Climate of Jessore" is the final outcome of the study. All available data have been used for preparing this climate; despite the fact that changes in climate are a long term continuous process and the scope of updating the climate remains unbolt. I am extremely grateful to Wg Cdr Md Shahidul Islam, OC 11 Sqn, for his all out support from the user point of view during the study. I am also grateful to Wg Cdr Ahmed Ali for his suggestion and guidance. I am thankful to Sgt Mokhlesur Rahman, Sgt Tuhin Ahmed and other Airmen and civ employees of Met Squadron, BAF MTR who helped in many ways while preparing this climate. The "Climate of Jessore" might help all concerns in BAF or any other organization for easy understanding the trend of weather throughout the year for their planning and execution of different events.

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November 2014

CLIMATE OF JESSORE

CHAPTER – I : INTRODUCTION

General Concept

1. Climate is commonly defined as the average pattern of variation in weather elements (temperature, humidity, atmospheric pressure, wind and precipitation) over a long period. The standard averaging period is 30 years as defined by World Meteorological Organization (WMO), but other periods may also be used depending on the purpose. Climate also includes statistics other than the average, such as the magnitudes of day-to-day or year-to-year variations. The difference between climate and weather is usefully summarized by the popular phrase "Climate is what you expect, weather is what you get". Climatology is nothing but the study of climate.

2. Weather forecasting for flight safety is the motto of any Met Sqn in BAF. Therefore, accurate forecasting is highly demanded from every corner for planning and accomplishment of different tasks vested upon them. Forecasting for different durations like, now-casting (up to 2 hrs) and very short range forecasting (up to 12 hrs) is important for daily flying; but short range (12 hrs to 72 hrs), medium range (72 hrs to 240 hrs), extended range (10 days to 30 days) and long range (30 days to 1 year) forecasting are very much important for planning of different occasions and sustained operations including flying effort planning. Now-casting and very short range forecasting is possible based on Persistence method[1] and Synoptic method[2] of forecasting; but the other types of forecasting depends on either climatology or Numerical Weather Prediction[3] (NWP). Again, in NWP method, different equations produce different results; so meteorologists must always use the other forecasting methods along with this one. Therefore, climatology has become an integral part of any forecasting.

3. Met Sqn, BAF MTR is a Main Meteorological Office (MMO) of BMD for Jessore area and is the only such BAF installation over southwestern part of the country. Historically, this office has acted as a hub for producing all meteorologists for BAF. For obvious reasons, BAF MTR generates the highest flying hours than any other base, and the Met Sqn plays a vital role in achieving that task. Met Sqn of BAF MTR is well equipped to collect and retain all the met data for not only daily forecasting but also long range forecast supported by valid climatological study. This study is aimed to generate the first ever climatological research for greater Jessore area using the Met data generated from this Sqn to act as future reference for all BAF flying and ceremonial activities, as well as for other relevant purposes. The Climate of Jessore is the study of total 30 years data for obtaining average conditions of different weather elements like temp, fog, thunderstorm/Nor'wester, heat wave, cold wave, rainfall, break monsoon etc. The contents of this study will certainly give an easy understanding about weather pattern of Jessore and adjoining area for the users of BAF for planning of flying effort and other occasions; or any other organization for their operational planning and execution.

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1. Persistence method assumes that what the weather is doing now, it will continue to do.

2. Synoptic method uses basic rules that the atmosphere follows ie, the observations, available charts, satellite image and radar observations etc.

3. In NWP method, forecasters take their observations and plug the numbers into complicated equations. Several ultra-high-speed computers run these various equations to make computer "models" which give a forecast for the next several days.

Questions for the Study

4. This study is the effort to search the pragmatic solutions of the following questions:

a. **Primary Question**. What are the climatic conditions in Jessore area at any time of a year?

b. **Secondary Question**. The primary question is further supplemented through the following secondary questions:

(1) What are the averages and extreme conditions of different weather elements?

(2) How long fog persists in winter, ie, what would be the usual flying start time in winter months?

(3) What would be the frequency and probable days for occurrence of Nor'wester in pre-monsoon season?

(4) What would be the heat wave and cold wave period in summer and winter?

(5) What would be the onset and withdrawal date of monsoon, period of continuous rain and break monsoon period during SW monsoon?

(6) How these climatic elements can significantly affect the flying, ceremonial and public life in Jessore area and when?

Limitations of the Study

5. Climatic study gets perfected by the duration of the study period, ie longer the time of study, more accurate the result likely to be. However, climate of any region is subject to change and evolvement for many reasons, For example, on 15 Sep 04, total 238mm rainfall occurred within one day and the amount of rainfall in 2004 (2477mm) was the highest rainfall within 30 years. On 05 Jun 84, total 242mm rainfall occurred within a day; as a result, the 30 years daily average rainfall became highest (20.2mm) in the month. In this way, sudden flash of very heavy rain in monsoon, effect of tropical cyclone, nor'wester etc affects the climate of a region. The weather and climate of Jessore area is also no exception. Therefore, this study should not be taken on the face value for any kind of 'perfect forecasting' in future abrasion, especially on now-casting.
The data regarding duration of fog has been taken for 20 years in this study instead of 30 years due to data constraints. There is scarcity of data preservation regarding occurrence of hailstorm over Jessore area. Therefore, the table regarding monthly hailstorm days may not be realistic. Though the climatological data have immense impact on any type of military operations and public life of a region, but here the findings have been postulated keeping in mind the priority in aviation.

Scope of Further Study

6. The 30 years raw data of different weather elements have been attested as annex, which may be utilized for further study. There is scope here for more study on the following issues:

a. Determine the days with low cloud amount 5 okta or more, when flying activities are hampered for low clouds.

b. Determine the days of poor visibility, when flying activities are hampered throughout the day.

c. Daily flying in BAF MTR is highly concern with FITS, particularly in pre-monsoon and monsoon season. Therefore, determine the time of high FITS (danger zone), when flying activities are stopped.

d. Determine the reason behind the high variability in yearly total amount of rainfall and rainy days found within the period of Jan1984 to Dec 2013.

Purpose of the Study

7. The purpose of this study is to provide comprehensive climatology of Jessore air field and adjoining area as a ready reference of a single source archive of relevant weather data. The reference can be readily used for planning and execution of flying syllabus, operational flying, ground operation, different events/ceremonies for saving time and effort with a pre-hand knowledge on different extreme weather conditions like High/low temperature, nor'wester, heavy rain, break monsoon etc.

Data Used

8. There are several elements that build up the weather and climate of a place. The major of these elements are five: temperature[1], pressure[2], wind[3], humidity[4], and precipitation[5]. Analysis of these elements within a longer time scale can provide the basis for defining the climate and forecasting weather. BAF Base Bir-Sreshto Matiur Rahman is acting as a Main Meteorological Office (MMO) of Bangladesh Meteorological Department (BMD) under a MOU between BAF and BMD since 1977. Therefore, Met Sqn, BAF MTR is having a data bank of about 35 years. In this study, total of 30 years real time data (from 01 January 1984 to 31 December 2013) of major three elements: temperature, wind, precipitation and its product: fog and thunderstorm/nor'wester have been taken into consideration. The round the clock data of those elements have been used in the following form:

a. Daily maximum temperature (^{0}C).

b. Daily minimum temperature (^{0}C).

c. Daily/Monthly no of rainy days.

d. Daily/Monthly total amount of rainfall (mm).

e. Monthly frequency of thunderstorm/nor'wester.

f. Time of occurrence of thunderstorm/nor'wester.

j. Monthly frequency of fog.

k. Time of cessation of fog.

l. Monthly maximum wind direction and speed (kts).

m. Monthly hailstorm days.

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1. Temperature is how hot or cold the atmosphere is, how many degrees it is above or below freezing. Temperature is a very important factor in determining the weather, because it influences or controls other elements of the weather, such as precipitation, humidity, clouds and atmospheric pressure.

2. Humidity is the amount of water vapor in the atmosphere.

3. Precipitation is any product of the condensation of atmospheric water vapour that falls under gravity. The main forms of precipitation include drizzle, rain, sleet, snow, graupel and hail. It may also include fog and mist.

4. Atmospheric pressure (or air pressure) is the weight of air resting on the earth's surface. Pressure is shown on a weather map, often called a synoptic map, with lines called isobars.

5. Wind is the movement of air masses, especially on the Earth's surface.

Data Processing and Inference

9. Geographical location of Bangladesh is from 20°34″ North Latitude to 26°38″ North Latitude. From 88°01″ East Longitude to 92°41″ East Longitude. Location of Jessore airfield is 23° 11′ 1″ N, 89° 9′ 39″ E and Elevation is 20ft Above Mean Sea Level (AMSL). The weather is extreme compared to other parts of the country. However, last 30 years data (01 Jan 1984 to 31 Dec 2013) of different weather elements are analysed here. The simple averaging method has been used throughout study. The way of processing the raw data of each element have been discussed and the inferences obtained from that processing are shown graphically. For better and easy understanding, the average conditions of weather elements like different types of recorded temperature, no of Nor'wester, time of occurrence of Nor'wester, thunderstorm days, no of rainy days, amount of rainfall, no of foggy days, pentad normal of rainfall amount, rainy days, temperature (max and min) etc are presented as graph/histogram.

Extreme Weather over Jessore

10. Some of the extreme weather elements recorded over Jessore air field within last 30 years (01 January 1984 to 31 December 2013) are listed below:

Ser No	Met Element	Value	Date/Time of Occurrence
1	Maximum temperature	43.2^{0}C	09 May 09
2	Minimum temperature	04.2^{0}C	09 JAN 13
3	Monthly highest mean maximum temperature	35.7^{0}C	In April
4	Monthly lowest mean minimum temperature	10.7^{0}C	In January
5	Maximum amount of rainfall in a day	255 mm	26 Sep 86
6	Maximum amount of rainfall in a month	856 mm	Sep 04
7	Maximum amount of rainfall in a year	2477mm	2004
8	Maximum no of rainy days in a month	27 days	Aug 06 & Jul 08
9	Maximum no of rainy days in a year	136 days	2013
10	Maximum no of thunderstorm in a month	23 days	Sep 97
11	Maximum no of nor'wester in a month	16 days	May 91
12	Maximum wind speed within 30 years	050/59kt	29 Nov 88

CHAPTER – II : PSYCHOTHERAPY OF TEMPERATURE

General Concept

1. The Temperature data have been considered for maximum and minimum only. The average conditions for monthly and fortnightly maximum and minimum temperature have been obtained by using simple averaging method. The recorded highest maximum and lowest minimum temperature in different months within last 30 years have been used directly in the graph. The analysis of temperature has been carried out to have the idea about the average maximum/minimum temperature condition throughout the year and the extreme conditions ever recorded. At first the monthly values are shown, thereafter, the fortnightly values are shown for better understanding about the change of temperature in extreme summer and winter conditions.

2. **Monthly Mean Maximum Temperature**. For obtaining monthly mean maximum temperature, the max temperature of each day in a month is summarized and divided by no of days of that month. It is the monthly mean max temp of one month of one year. Thereafter, 30 years mean is obtained in the same way, which are added thereafter and divided by 30 for obtaining monthly mean maximum temperature. The obtained values are used in the graph.

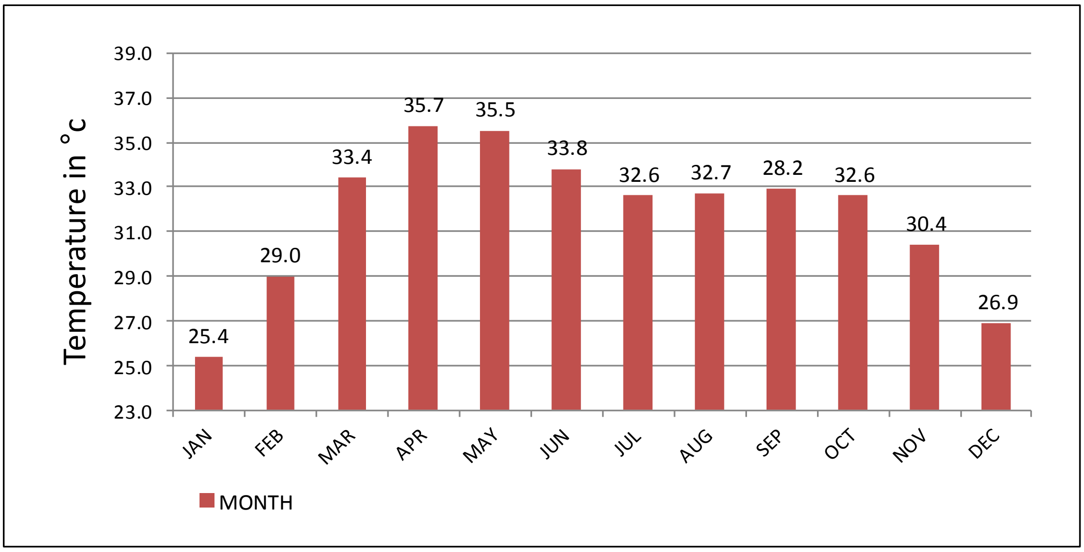

Fig-1 : Monthly mean maximum temperature for the period of 1984 to 2013

3. **Analysis**. The above graph (Fig-1) reveals that the month April and May is having mean max temp 35°c - 36°c, March and June is having mean max temp 33°c - 34°c, July to October is having mean max temp 32°c - 33°c and other months mean max temp is 30°c or below. Therefore, the month April is the hottest month of the year and the next hottest month is the May. The max temp is almost similar in the months July - October and March. Thereafter, max temp starts falling and almost similar max temp prevail in the month February and November. The max temp in the month January is the lowest.

4. **Monthly Highest Maximum Temperature**. The highest max temp obtained within 30 years (01 Jan 84 to 31 Dec 13) for each month is used directly in the graph. This graph gives the idea about the recorded highest maximum temp within last 30 years.

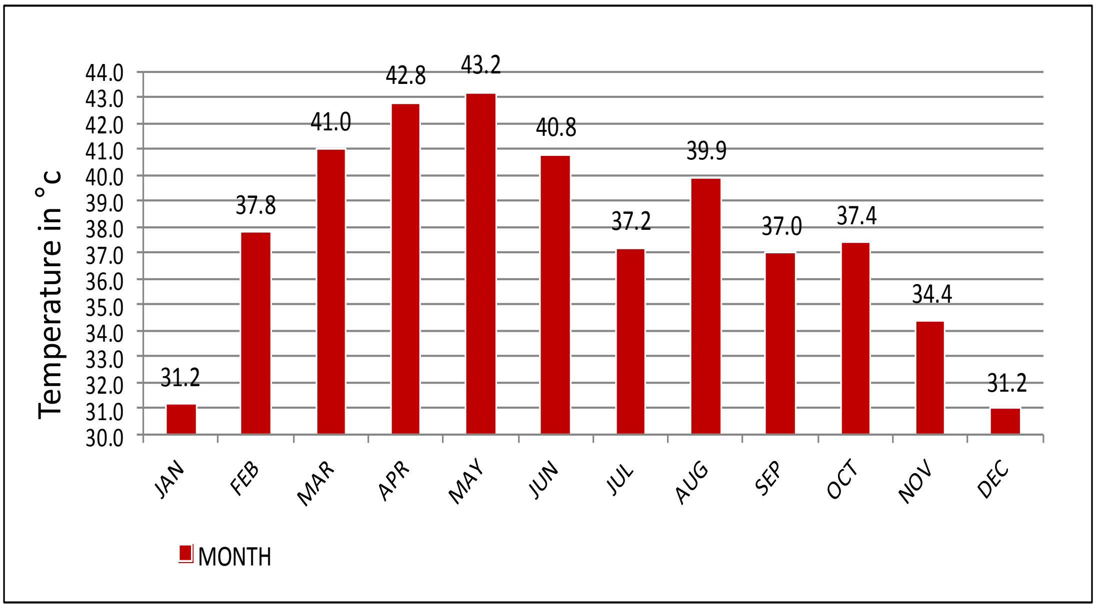

Fig-2 : Monthly highest maximum temperature for the period of 1984 to 2013

5. **Analysis**. It is revealed from this graph (Fig-2) that highest max temp was obtained in the month of May (43.2ºc; 09 May 09) and the next max temp was obtained in April (42.8ºc; 27 April 09). The temp in July (37.2ºc) is less than the temp in August (39.9ºc); it may be due to the more no rainy days and cloudy sky in July when maximum temp cannot rise. The temp in August (39.9ºc) is more because of break in monsoon, when cloud decreases and rainfall ceases. The temp in July (37.2ºc) is almost similar to the temp of the month of February (37.8ºc), September (37.0ºc) and October (37.4ºc). The max temp obtained in the month December and January (31.2ºc) was similar.

6. **Monthly Mean Minimum Temperature**. For obtaining monthly mean minimum temperature, the minimum temperature of each day in a month is summarized and divided by no of days of that month. It is the monthly mean minimum temp of one month of one year. Thereafter, 30 years mean is obtained in the same way, which are added and divided by 30 for obtaining monthly mean minimum temperature and the values are used in the graph.

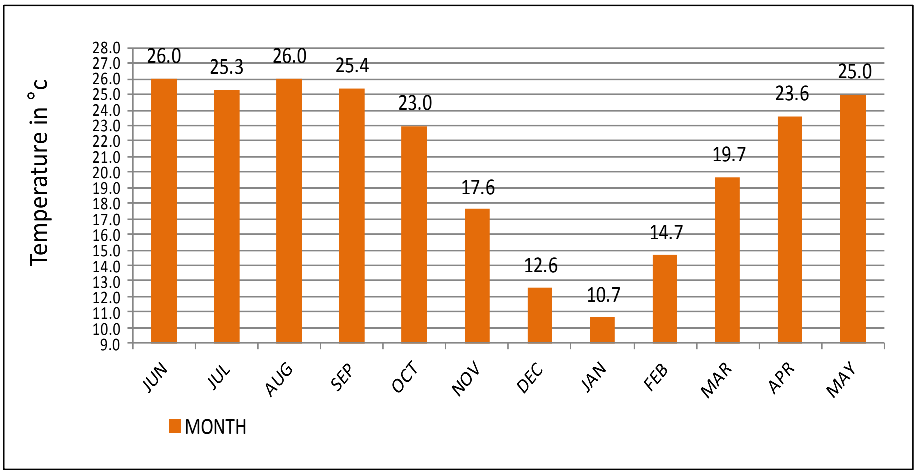

Fig-3 : Monthly mean minimum temperature for the period of 1984 to 2013

7. **Analysis**. It is revealed from the graph (Fig-3) that the month May to September is having mean minimum temp within 25°c - 26°c, the month April & October is having the mean min temp within 23°c - 24ºc. The temp starts falling sharply from November till January and January is having mean minimum temp 10.7°c. Therefore, January is the coldest month of the year. The temp starts rising sharply from January till June.

8. **Monthly Lowest Minimum Temperature**. The lowest minimum temperature obtained within last 30 years (01 Jan 84 to 31 Dec 13) for each month are used directly in the graph.

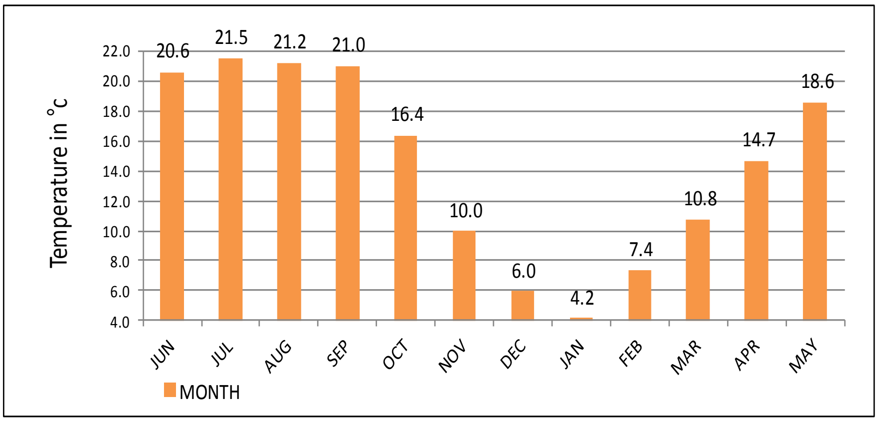

Fig-4 : Monthly lowest minimum temperature for the period of 1984 to 2013

9. **Analysis**. It is obtained from this graph (Fig-4) that the lowest minimum temp within last 30 years was recorded in the month of January (4.2ºc on 09 Jan 13). The lowest temp in the month of June to September was almost similar (within 20.6ºc to 21.5ºc). The temp starts falling sharply from October till January and thereafter rising sharply till July.

10. **<u>Fortnightly Mean Max Temp</u>**. For obtaining fortnightly mean maximum temperature, the max temperature of each day in a fortnight is summarized and divided by no of days (14). It is the fortnightly mean max temp of one fortnight of one year. Thereafter, 30 years mean is obtained in the same way, which are added and divided by 30 for obtaining fortnightly mean maximum temperature and the values have been used in the graph.

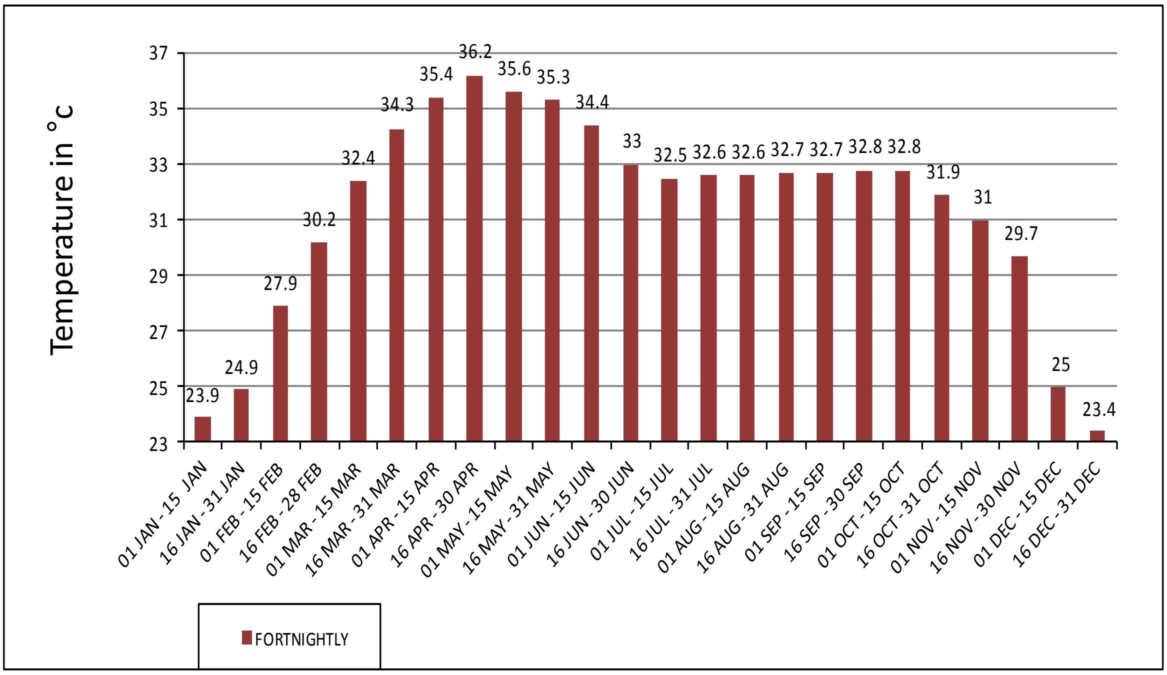

Fig-5 : Fortnightly mean maximum temp for the period of 1984 to 2013

11. **<u>Analysis</u>**. Though the monthly mean max temp graph (Fig-1) shows that the temp in April is highest and value is 35.7°c but this graph (Fig-5) show that the 2nd fortnight of April (16 April to 30 April) is having maximum average temp of the year and the value is 36.2°c. Therefore, the **2nd fortnight of April (16 April to 30 April) is the hottest period of the year**. The 2nd fortnight of December (16 Dec to 31 Dec) is having the lowest mean max temp (23.4°c), thereafter it starts rising gradually till 30 April. From 01 January to 31 January the max temp rises by 1°c. From 31 January to 31 March it increases with a rate of 2°c - 3°c per fortnight (14 days); thereafter it increases up to 30 April with a rate of 1°c - 2°c per fortnight. The mean max temp starts falling slowly from 1st fortnight of May till 2nd fortnight of Jun and it remain almost similar (within 32°c - 33°c) till 1st fortnight of Oct (01 Oct to 15 Oct). Thereafter, mean max temp starts falling slowly till 2nd fortnight of November. In the 1st and 2nd fortnight of December, temp fall is sharp and the difference between min and max temp decreases that time. Therefore, **people really fill cold after 30th November.** During heat wave conditions, flying becomes restricted, people become sick due to heat stroke and human performance reduces due to sweating.

12. **Fortnightly Mean Minimum Temp**. For obtaining fortnightly mean minimum temperature, the minimum temperature of each day in a fortnight is summarized and divided by no of days (14). It is the fortnightly mean minimum temp of one fortnight of one year. Thereafter, 30 years mean is obtained in the same way, which are added and divided by 30 for obtaining fortnightly mean minimum temperature and the values have been used in the graph.

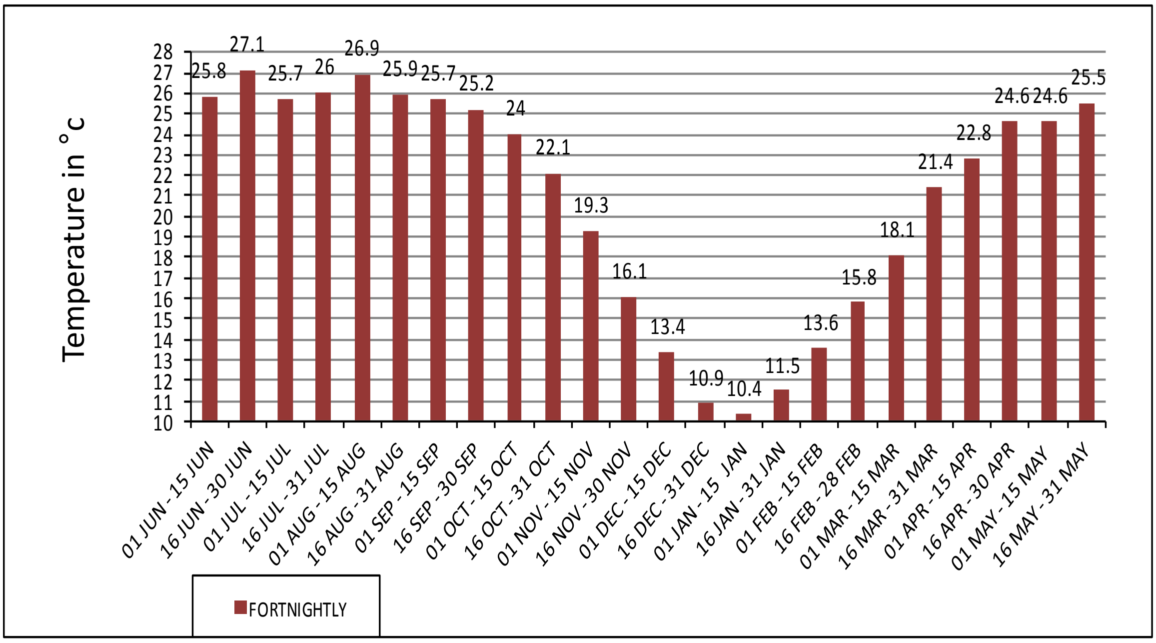

Fig-6 : Fortnightly mean minimum temp for the period of 1984 to 2013

13. **Analysis**. It is revealed from this graph (Fig-6) that the 1st fortnight of January is the coldest period of the year with the mean temp 10.4°c. The mean minimum temp remain above 24ºc from 2nd fortnight of April to 1st fortnight of October. Thereafter, temp starts falling 2ºc - 3ºc per fortnight till 1st fortnight of January. The temp starts rising from 2nd fortnight of January till 1st fortnight of April at the rate of 2-3ºc per fortnight. Most of the cold wave occurs in the 1st fortnight of January. A cold wave can cause death and injury to livestock and wildlife. Exposure to cold mandates greater caloric intake for all animals, including humans. The belief that more deaths are caused by cold weather in comparison to hot weather is true as a result of cold, flu, pneumonia, etc.

14. **Fortnightly Highest Max Temp**. The highest maximum temp obtained within last 30 years (01 Jan 84 to 31 Dec 13) for each Fortnight have been used directly in the graph.

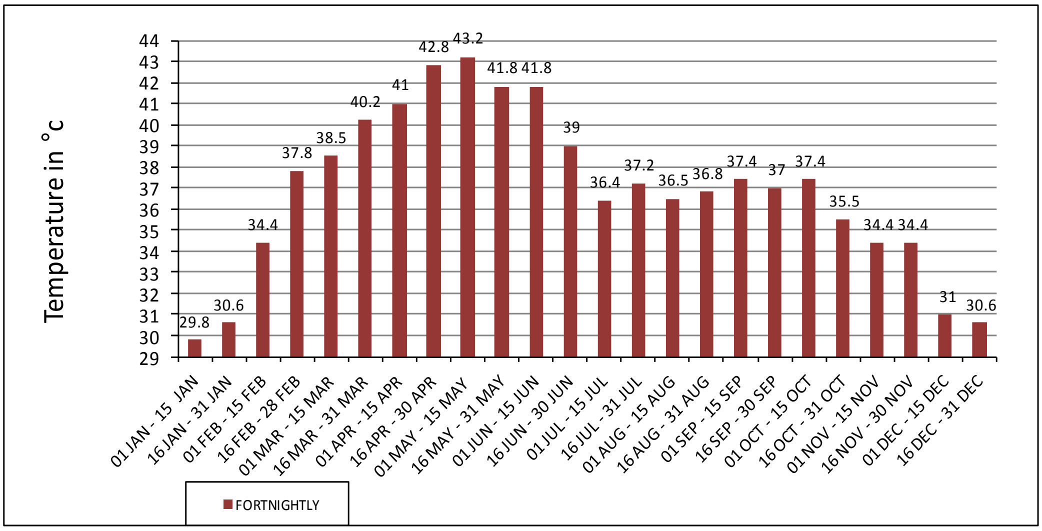

Fig-7 : Fortnightly highest maximum temp for the period of 1984 to 2013

15. **Analysis**. This graph (Fig-7) reveals that the highest maximum temp within last 30 years was obtained in the 1st fortnight of May and the lowest maximum temp was obtained in the 1st fortnight of January. The temp in 1st fortnight of July is less than 2nd fortnight of June and July. It may be due to more no of rainy days and cloudy sky in that period.

16. **Fortnightly Lowest Minimum Temp**. The lowest minimum temperature obtained within last 30 years (01 Jan 84 to 31 Dec 13) for each Fortnight have been used directly in the graph.

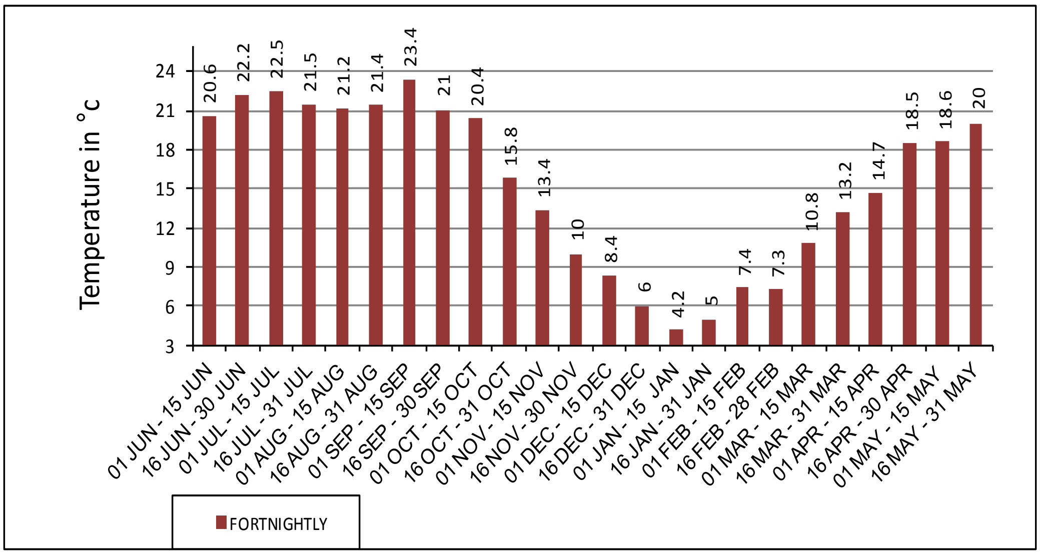

Fig-8 : Fortnightly lowest minimum temp for the period of 1984 to 2013

17. **Analysis**. This graph shows that the lowest temp within last 30 years obtained 4.2°c in the 1st fortnight of January. (Fig-8)

18. **<u>Monthly Average no of Days Having Temperature ≤ 10ºc</u>**. For obtaining this value, the total no of days having temp ≤10ºc in each month is summarized for 30 years and then divided by 30.

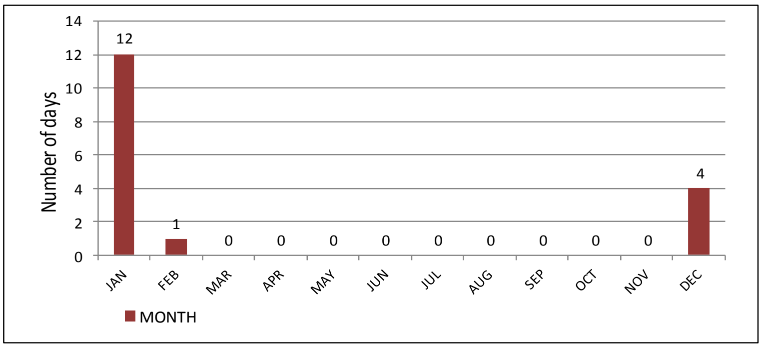

Fig-9: Monthly average no of days having temperature ≤ 10ºc for the period of 1984 to 2013

19. **<u>Analysis</u>**. This graph (Fig- 9) shows that the days having temp ≤10ºc are more in January. The average no of days are 4 days in December (max 13 days in 2007 & 2010), 12 in January (max 23 days in 2003) and 1.3 days in February (max 08 days in 2008). The study reveals that January is the coldest month of the year, then December. It is revealed from Fig-6 that the 1st fortnight of January (01 Jan to 15 Jan) is having average temp 10.4ºc, which is the coldest period of the year. Therefore, most cold wave occurs in January and it is mostly in the 1st fortnight of January. The **maximum no of moderate (6-8ºc) to severe (4-6ºc) cold wave was found in 1st fortnight of January (12 cold wave),** then in 2nd fortnight (09 cold wave); cold wave also found in 2nd fortnight of December (04 cold wave).

20. **<u>Monthly Average no of Days Having Temperature ≥38^{0}c</u>**. For obtaining this value, the total no of days having temp ≥38ºc in each month is summarized for 30 years and then divided by 30.

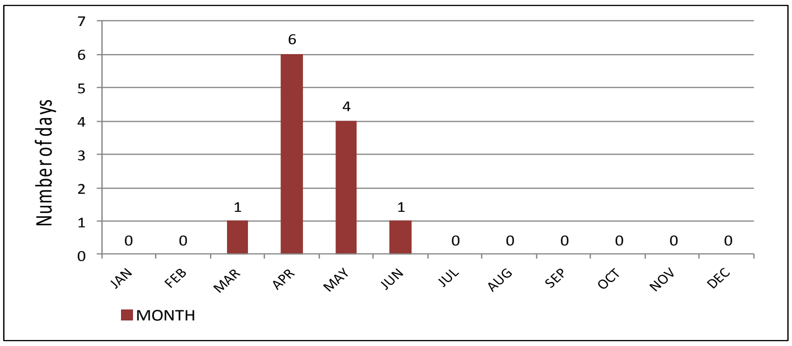

Fig-10 : Monthly average no of days having temperature ≥ 38 ^{0}C for the period of 1984 to 2013

21. **<u>Analysis</u>**. This graph (Fig-10) shows that the days are 06 in April (max 20 days in 1992), 04 in May (Max 14 days in 2004), 0.8 day in March (max 06 days in 1995) and 1.4 days June (max 12 days in 2005). Therefore, it is revealed from the graph that, April is the hottest month of the year and then May. It is revealed from Fig-5 that the 2nd fortnight of April is having the highest average maximum temp (36.2ºc) of the year and 1st fortnight of May is the 2nd highest temp (35.6ºc). Therefore, most of the **severe (temp 40-42ºc) heat waves were found in the 2nd fortnight of April** (11 heat wave), then the 1st fortnight of May (05 heat wave); heat wave also found in the 1st fortnight of April (02 heat wave), 2nd fortnight of May (04 heat wave) and 1st fortnight of June (01 heat wave). Thus, Daily flying and public life would be affected by cold wave in January and heat wave in April and May.

CHAPTER – III : FOG INVESTIGATION

General Concept

1. The data for monthly foggy days have been considered for the period from 01 Jan 84 to 31 Dec 13 (30 years). The duration of fog data have been considered for the period from 01Jan 93 to 31 Dec 12 (20 years). The monthly average cessation time of Fog has been obtained for the month November to March. As because, the foggy days in other months (Apr, Sep and Oct) are very less in number and short lived (ceases within 0800hrs).

2. **Monthly Average no of Foggy Days**. The total no of foggy days in each month is summarized for 30 years and divided by 30 for obtaining monthly average no of foggy days and these values are in the graph.

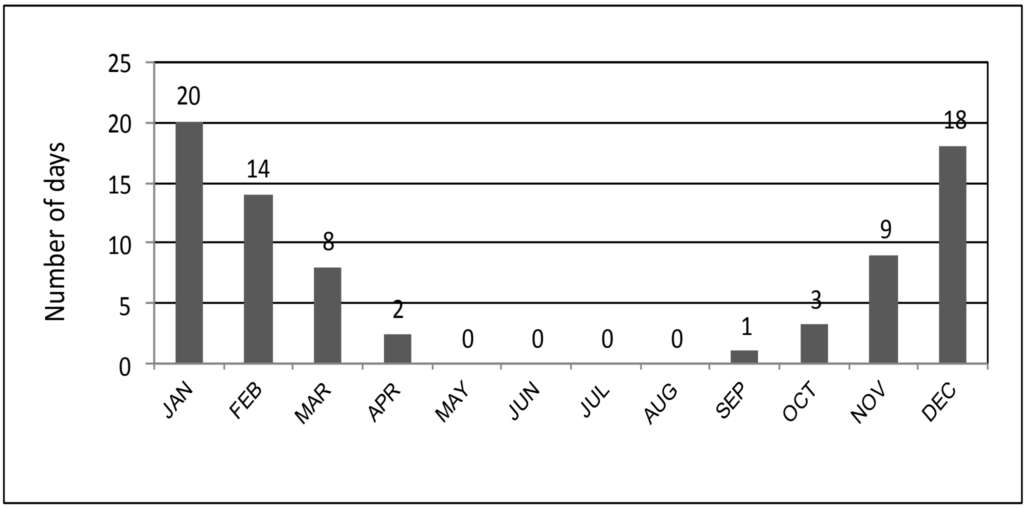

Fig-11 : Monthly average no of foggy days for the period of 1984 to 2013

3. **Analysis**. The study of this graph (Fig-11) reveals that fog starts in September; no of days increases gradually till January and no days decreases gradually from February to April. Maximum no of foggy days are in January; it is average 20 days. The average foggy days from September to April are 01, 03, 09, 18, 20, 14, 08 and 02 days respectively.

The Persistence of Fog in Different Months

4. **Average Persistence of Fog in November**. The values are obtained by averaging the cease time of foggy days in each month for last 20 years (1993 to 2012). The cease time has been rounded up in the next hour when the cease time passes 30 minutes of the hour; otherwise it is rounded up in the previous hour.

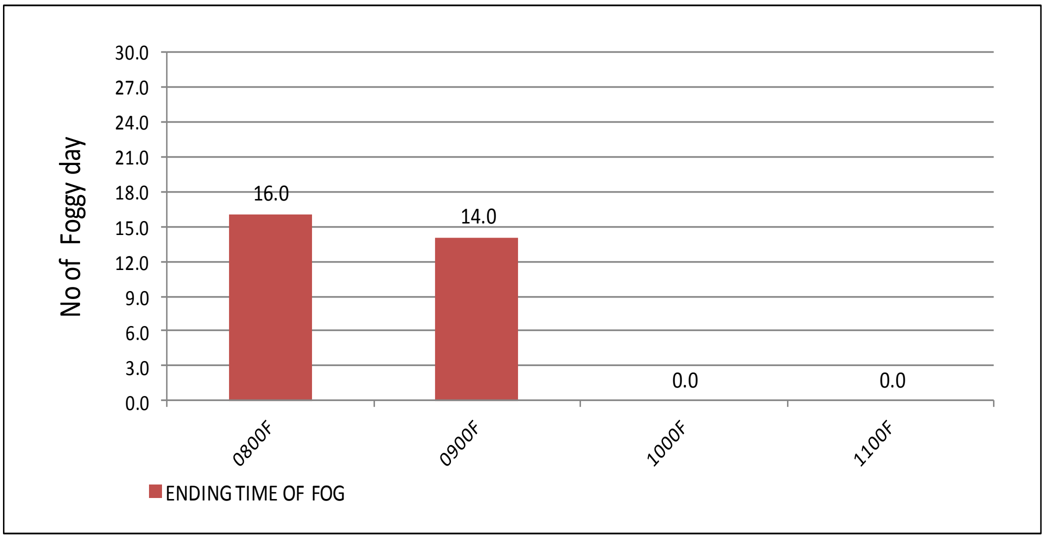

Fig-12 : Average end time of fog in November for the period of 1993 to 2012

5. **Analysis**. It is revealed from the study of the graph (Fig-12) that, the Fog ceased within 0800F for 16 days and 0900F for 14 days in November. It means 53% cases fog persists up to 0800F and 47% cases up to 0900F. The average foggy days in November was 09 days (from Fig-11). Therefore, in November, average 05 days fog persists up to 0800F and 04 days up to 0900F. Therefore, in November, most of the days flying would be started after 0800F.

6. **Average Persistence of Fog in December**. In December fog occurs average 18 days and it persists more time than November.

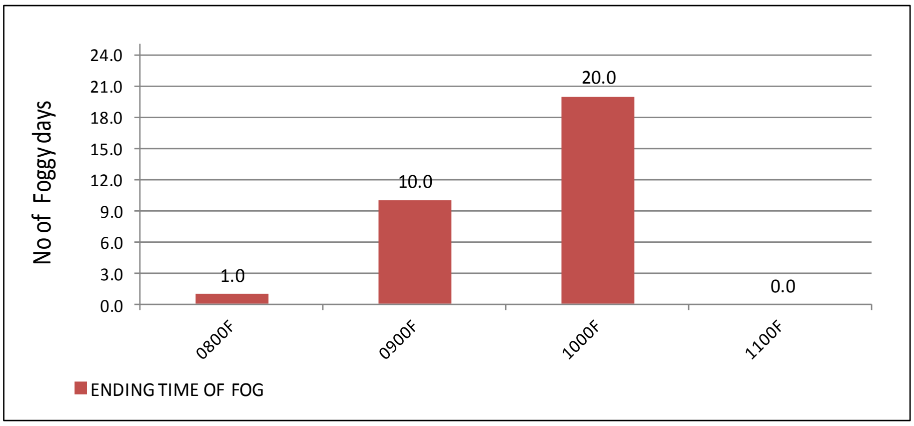

Fig-13 : Average end time of fog in December for the period of 1993 to 2012

7. The graph (Fig-13) shows that, the fog persists 20 days up to 1000F, 10 days up to 0900F and only one day up to 0800F within a month. It means, for 65% cases fog persists up to 1000F, for 32% cases up to 0900F and for 3% cases up to 0800F. The average foggy days in December was 18 days (from Fig-11). Therefore, in December, average 12 days fog persists up to 1000F, 5.75 days up to 0900F and only 0.25 days up to 0800F. Therefore, in December, flying would not be possible before 1000F.

8. **Average Persistence of Fog in January**. In January, average foggy days are 20 (Fig-11), which is the highest in the year and the persistence time also more than any other months.

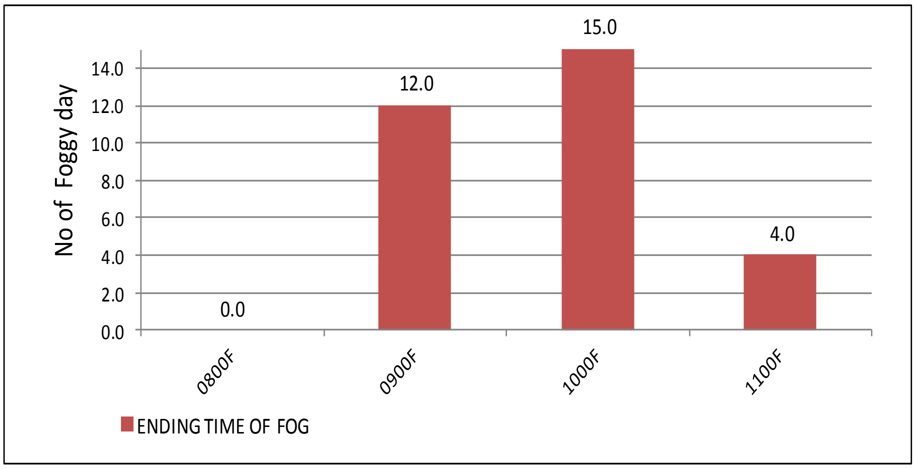

Fig-14 : Average end time of fog in January for the period of 1993 to 2012

9. The graph (Fig-14) shows that the fog persists up to 0900F for 12 days, up to 1000F for 15 days and up to 1100F for 04 days in a month. It means, for 40% cases fog persists up to 0900F, for 50% cases up to 1000F and for 13% cases up to 1100F. The average foggy days in January was 20 days. Therefore, in January, average 08 days fog persists up to 0900F, 10 days up to 1000F and 02 days up to 1100F. In January, most of the days flying may be started after 1000F and some days after 1100F.

9. **Average Persistence of Fog in February**. In an average, 14 days fog occurs in February (Fig-11) and the persistence time also reduces than January.

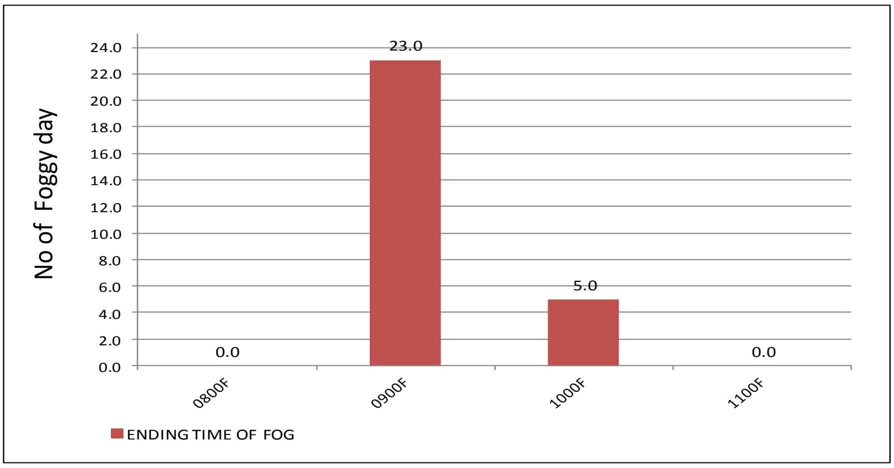

Fig-15 : Average end time of fog in February for the period of 1993 to 2012

11. The graph (Fig-15) shows that the fog in February persists up to 0900F for 23 days and up to 1000F for 05 days. It means, for 82% cases fog persists up to 0900F and for 18% cases up to 1000F. The average foggy days in February was 14 days. Therefore, in February, average 11.5 days fog persists up to 0900F and 2.5 days up to 1000F.

12. **Average Persistence of Fog in March**. In March, no of foggy days reduces but the time of persistence remain almost same as February. Average foggy days are 08 in the month (Fig-11).

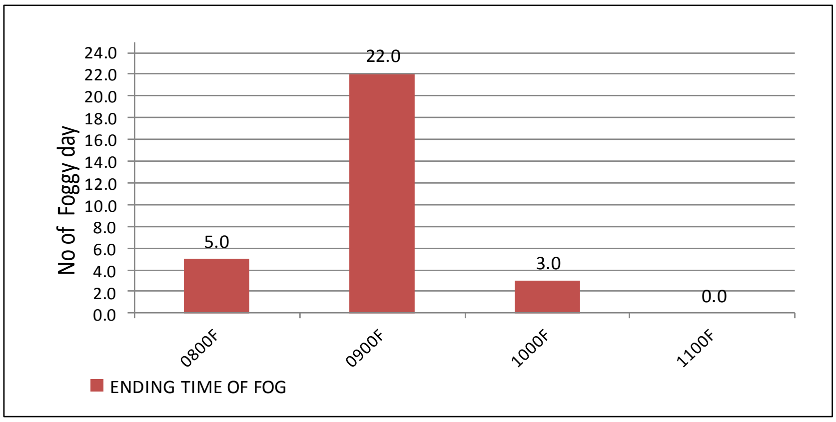

Fig-16 : Average end time of fog in March for the period of 1993 to 2012

13. The graph (Fig-16) shows that the fog persists up to 0800F for 05 days, up to 0900F for 22 days and up to 1000F for 03 days in a month. It means, for 17% cases fog persists up to 0800F, for 73% cases up to 0900F and for 10% cases up to 1000F. The average foggy days in March was 08 days. Therefore, in March, average 1.3 days fog persists up to 0800F, 06 days up to 0900F and 0.7 days up to 1000F. Thus, Flying would be started after 0900F in most of the days in February and March.

CHAPTER – IV : THUNDERSTORM BEHAVIOUR

General Concept

1. A thunderstorm is the most hazardous weather as far as flying is concern. It occurs almost all month of the year except Nov, Dec and Jan. The pre-monsoon thunderstorms occur from Feb to May (till monsoon set) are the most devastating and these thunderstorms are also called Nor'wester or Kalbaishakhy in Bengali. Thundestorms in other months though are more in no but those are monsoon or post-monsoon thunderstorms and not that much devastating. In this study, the pre-monsoon thunderstorms which had wind speed ≥22 Kts, only those have been considered as Nor'wester. The month-wise total no of Nor'wester occurred within last 30 years have been also shown in this chapter for understanding the frequency. During analysis, it was found that some nor'wester occurred with rain, some nor'wester occurred without rain and some cases considerable amount of rain occurred but wind speed did not cross 22 Kts; so it was not counted as nor'wester. Therefore, for finding out the susceptible days of nor'wester, amount of rain and no of rainy days in different months from "Precipitation Study" chapter have also been taken into consideration. The monthly and yearly maximum wind speed is also shown in this chapter, because the most destructive power of Nor'wester is the wind. The monthly or yearly maximum wind speed in pre-monsoon months is mostly associated with Nor'wester, but in the post-monsoon, it is mostly associated with Tropical Cyclone. The monthly maximum wind speed and direction for last 30 years is shown in table-1.

2. **Monthly Average no of Thunderstorm Days**. This value is obtained by averaging the no of thunderstorm days in each month for last 30 years.

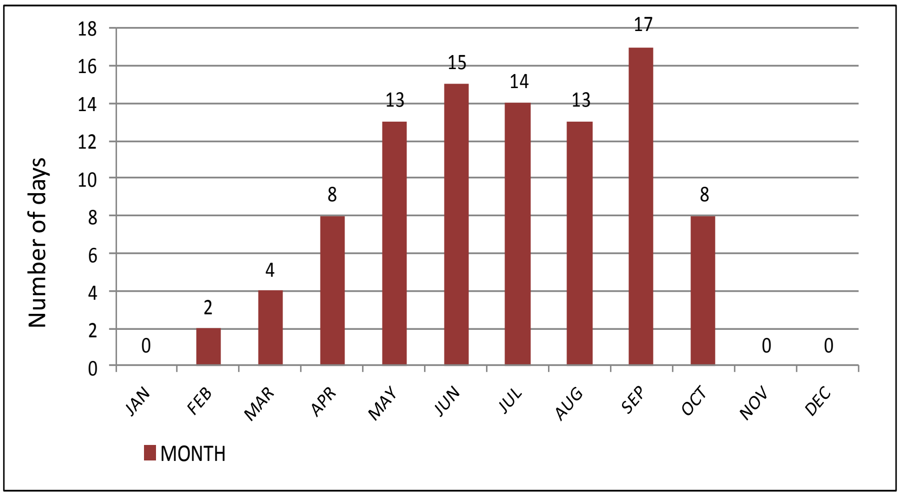

Fig-17 : Monthly average no of thunderstorm days for the period of 1984 to 2013

3. **Analysis**. It is revealed from this graph (Fig- 17) that the thunderstorm occurs from February to October and maximum no of Thunderstorm occurs in September (17 days). The average no of Thunderstorms occurs 02 days in February, 04 days in March, 08 days in April, 13 days in May, 15 days in June, 14 days in July, 13 days in August and 08 days in October. The thunderstorms occur in pre-monsoon season (February to May), most of them are associated with Nor'wester but some of them may not be Nor'wester.

4. **Monthly Average No of Norwester**. This value is obtained by averaging the no of Nor'wester (wind speed 22 kts or more) in each month for 30 years. Nor'wester occurs in pre-monsoon season only; thus, the month February to May is being taken into consideration.

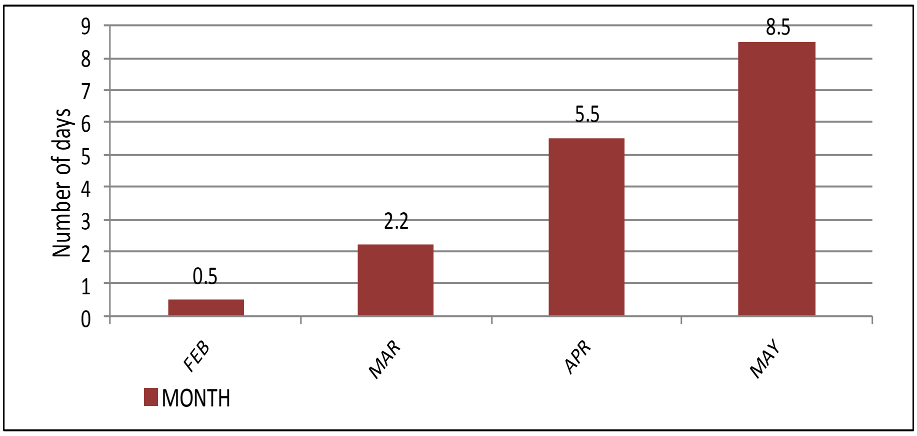

Fig-18 : Monthly average no of norwester for the period of 1984 to 2013

5. **Analysis**. The study reveals that the frequency of Nor'wester increases gradually with the progress of month from February to May. The average no of Nor'wester occurs 0.5 in February, 2.2 in March, 5.5 in April and 8.5 in May. The occurrences of Nor'wester continue till onset of Monsoon; thus, Nor'wester may occur in the 1st week of June also. (Fig- 18)

6. **Time of Occurrence of Norwester**. This graph shows the time of occurrence of total no of Nor'wester within last 30 years (01 Jan 84 to 31 Dec 13).

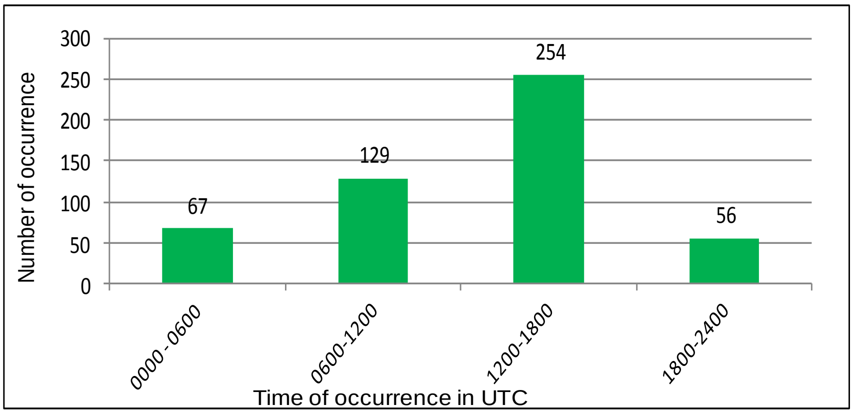

Fig-19 : Time of occurrence of norwester for the period of 1984 to 2013

7. **Analysis**. The study of this graph (Fig-19) reveals that the maximum no of Nor'wester (total 254) occurred within the time 1200Z - 1800Z, total 129 Nor'wester occurred within 0600Z - 1200Z, total 67 Nor'wester occurred within 0000Z - 0600Z and total 56 Nor'wester occurred within 1800Z - 2400Z. Therefore, about 50% Nor'wester occurred within 1200Z - 1800Z, 25% Nor'wester occurred within 0600Z - 1200Z, 14% Nor'wester occurred within 0000Z - 0600Z and only 11% Nor'wester occurred within 1800Z - 2400Z. Therefore, any outside activity including flying, driving or any public gathering/ceremony would be affected by nor'wester largely within 1200Z – 1800Z.

Month-wise Total No of Nor'wester

8. **Total no of Norwester in February**. This graph shows the total no of Nor'wester occurred in February within last 30 years. Total 18 Nor'wester occurred.

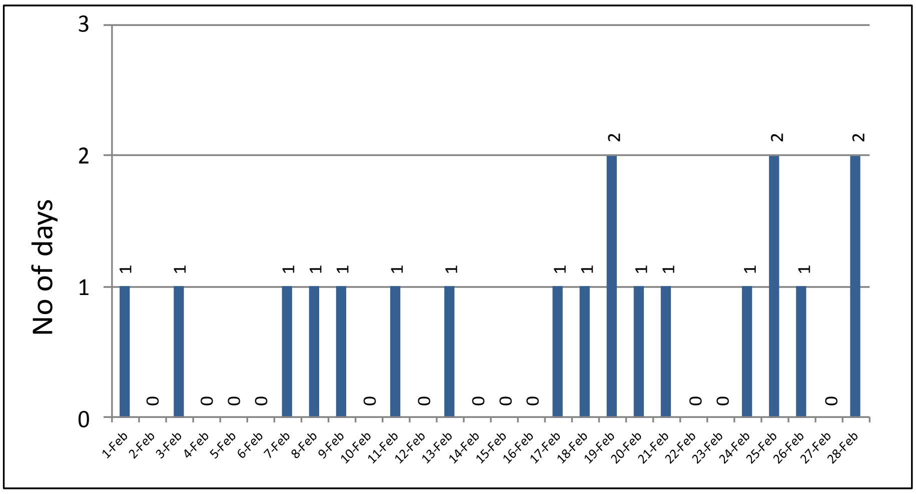

Fig- 20 : Total No of nor'wester (wind speed ≥22 kts) for the period of 1984 to 2013

8a. **Analysis**. The months of February to May are known as the pre-monsoon season, as well as the Nor'wester period for Bangladesh. Therefore, the rainfall occurs within this period are mostly associated with Nor'wester. The study of this graph (Fig-20) doesn't give any indication regarding vulnerability of nor'wester day. However, probability of occurrence of nor'wester on 19^{th}, 25^{th} and 28^{th} February is more than any other day (probability <10%).

9. **Total No of Norwester in March**. This graph shows the total no of Nor'wester occurred in March within last 30 years. Total 68 Nor'wester occurred.

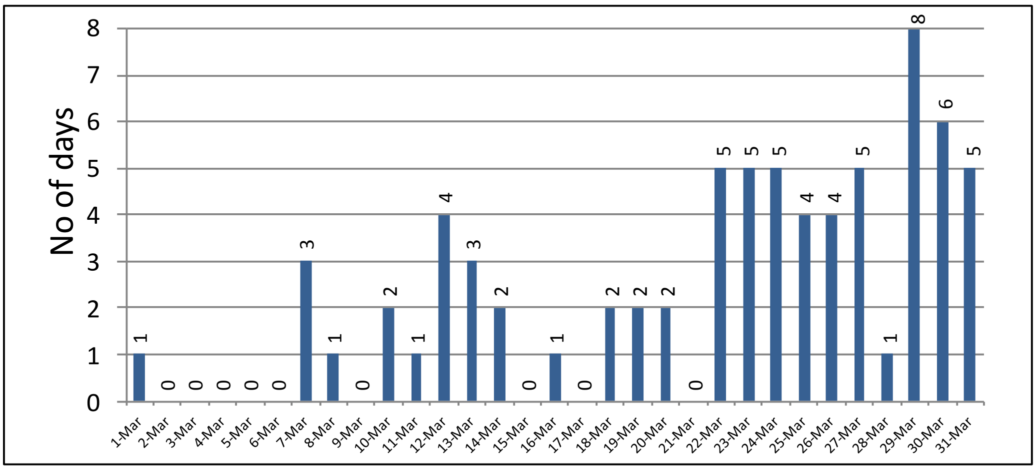

Fig-21: No of norwester (wind speed ≥22 kts) in March for the period of 1984 to 2013

9a. **Analysis**. The frequency of nor'wester in March is 2.2. The study of this graph (Fig-21) reveals that the susceptible days for nor'wester may be 29^{th} and 30^{th} March, where the probability is 20% and 27% respectively. As the frequency of nor'wester is less in March, the maximum probable (20% and 27%) days have been considered as susceptible days.

10. **Total No of Norwester Days in April**. The graph shows the total no of Nor'wester occurred in April within last 30 years. Total 161 Nor'wester occurred.

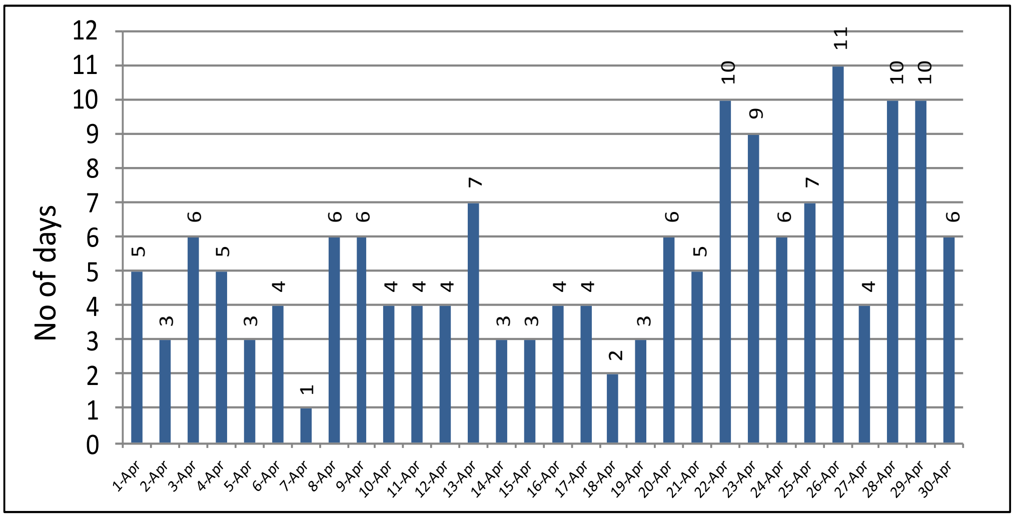

Fig-22: No of norwester (wind speed ≥22 kts) in April for the period of 1984 to 2013

10a. **Analysis**. The frequency of nor'wester in April is 5.5. The study of the graph (Fig-22) reveals that the susceptible days for nor'wester are 22nd, 23rd, 26th, 28th and 29th April and the probability is around 30%.

11. **Total No of Nor'wester Days in May**. This graph shows the total no of Nor'wester occurred in each year within last 30 years. Total 256 Nor'wester occurred in May.

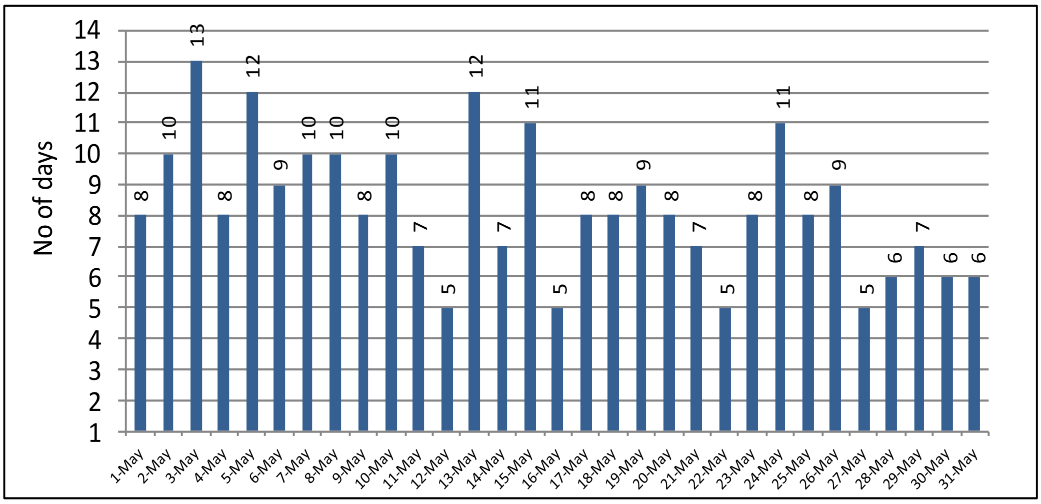

Fig-23 : No of norwester (wind speed ≥22 kts) in May for the period of 1984 to 2013

11a. **Analysis**. The frequency of nor'ester in May was 8.5. The study of this graph (Fig-23) reveals that the susceptible days of nor'wester may be 2nd, 3rd, 5th, 7th, 8th, 10th, 13th, 15th and 24th May and the probability is 30% or more. However, the frequency of nor'wester increases gradually from February to May and the intensity decreases. Therefore, for finding out the susceptible days of nor'wester, amount of rain and no of rainy days in all these months from "Precipitation Study" chapter have also to be taken into consideration.

Maximum Wind Speed

12. **Monthly Maximum Wind Speed**. The maximum wind speeds found in each month within last 30 years have been used directly in this graph.

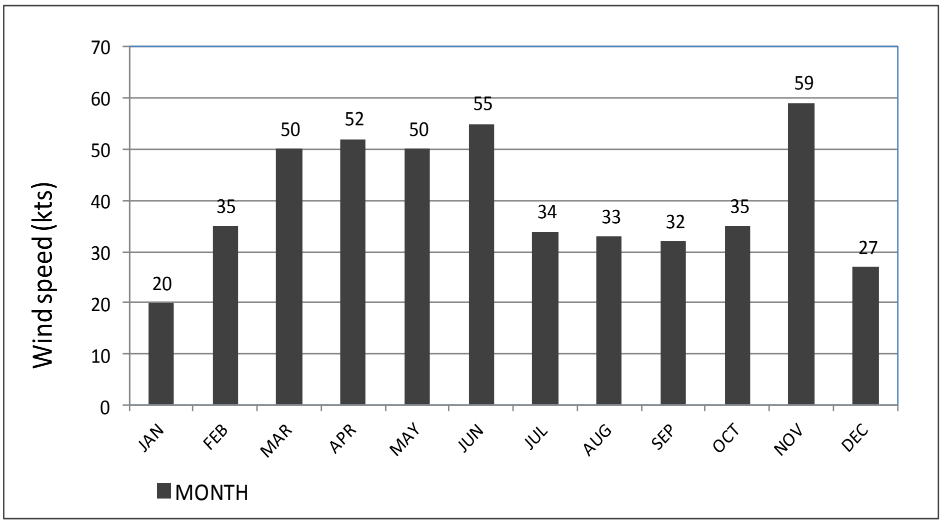

Fig-24 : Monthly maximum wind speed for the period of 1984 to 2013

13. **Analysis**. It is revealed from the study of this graph (Fig-24) that the maximum wind speed within last 30 years was 59 Kts in November. As November is post-monsoon season, thus the wind speed was associated with Tropical Cyclone. The wind speeds in Mar to Jun are more than 50 Kts, which are associated with Nor'wester.

14. **Yearly Maximum Wind Speed**. This graph (Fig-25) shows the maximum wind speed found in each year within the period of 1984 to 2013. It is revealed that the maximum wind speed within last 30 years was found 59 Kts from Nov 1988 Cyclone.

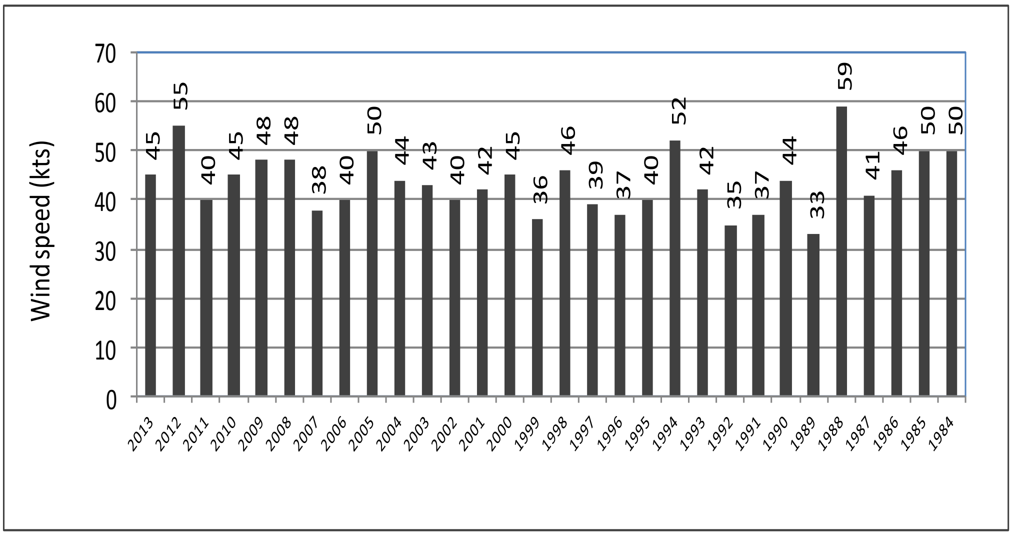

Fig-25 : Yearly maximum wind speed for the period of 1984 to 2013

CHAPTER – V : PRICIPITATION STUDY

General Concept

1. The rainfall data have been considered in the form of rainfall amount and no of rainy days for the period of 01 Jan 1984 to 31 Dec 2013. The data have been analyzed in different ways like monthly average rainfall amount and rainy days; pentad (five days) average rainfall amount and rainy days; and daily average rainfall amount and rainy days. The yearly total amount of rainfall and total no of rainy days within last 30 years have also been shown in this chapter. The pentad (five days) average rainfall is a world-wide used meteorological terminology for determining the establishment of rainy season, rainfall variability etc. Therefore, the pentad has been considered from the month April to October only as the monsoon season is from June to September for Bangladesh. The Daily rainfall data has been considered from the month February to October; because, the amount of rainfall and rainy days in other months (November to January) are not significant. Moreover, the rainfall from November to January is mostly associated with either Tropical Cyclone or western disturbances which have not been considered here.

Monthly Average Rainfall Amount and Rainy Days

2. **Monthly Average Amount of Rainfall**. These values are obtained from last 30 years average amount of rainfall for each month.

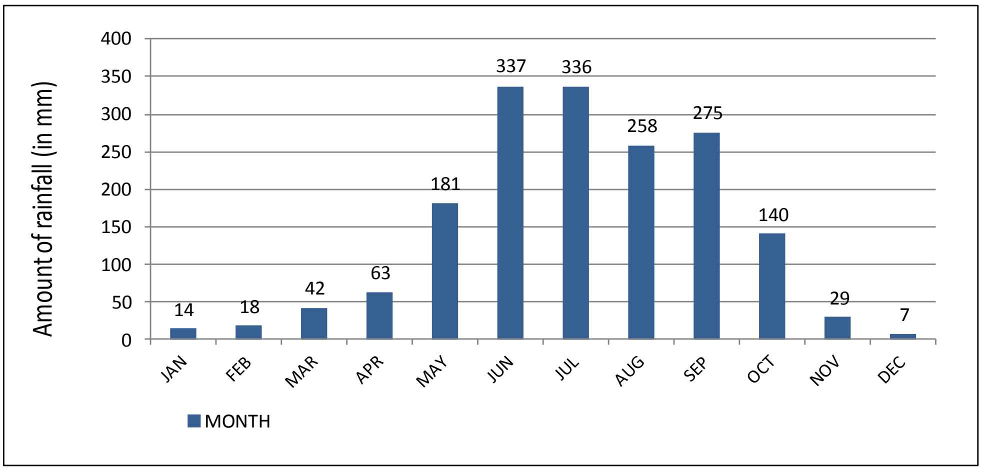

Fig-26 : Monthly average amount of rainfall for the period of 1984 to 2013

3. **Analysis**. The graph (Fig- 26) shows that the maximum amount of rain occurs in June & July, 337mm & 336mm respectively. In August, average rainfall amount (258mm) decreases compare with last two months but in September (275mm) again it increases. The amount of rainfall decreases sharply from October (140 mm) till December (07 mm). The monthly total amounts of rainfall within last 30 years have been shown in Table-2.

4. **Monthly Average No of Rainy Days**. This is the average no of rainy days of each month for last 30 years. In this calculation, 1mm rainfall in a day has been considered as a rainy day.

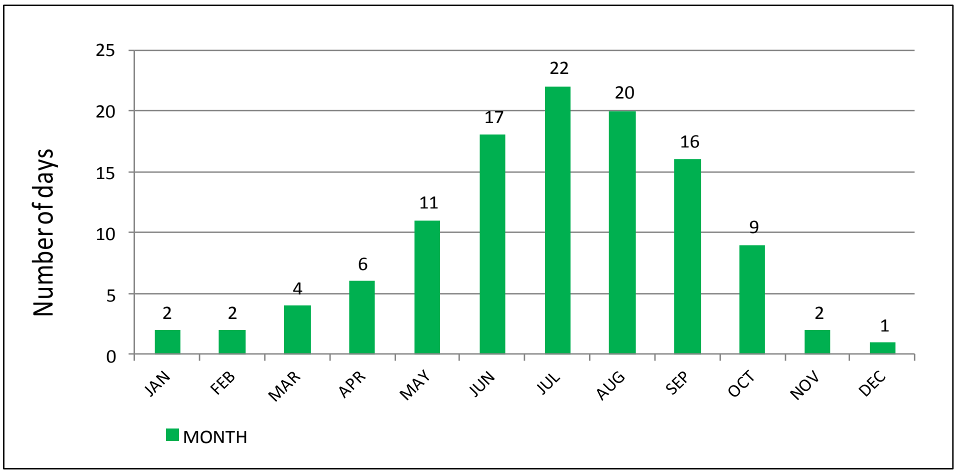

Fig-27 : Monthly average no of rainy days for the period of 1984 to 2013

5. **Analysis**. It is revealed from this graph that, the rainfall may occur in every month of the year and the maximum average rainy days are in July & August, 22 & 20 days respectively (Fig- 27). Yearly total no of rainy days for 30 years are shown in graph (Fig-29). The monthly total no of rainy days within last 30 years are shown in Table-3.

Yearly Amount of Rainfall and Rainy Days

6. **Yearly Total Amount of Rainfall**. The yearly total amount of rainfall for the period of Jan 1983 to Dec 2013 has been used directly in this graph.

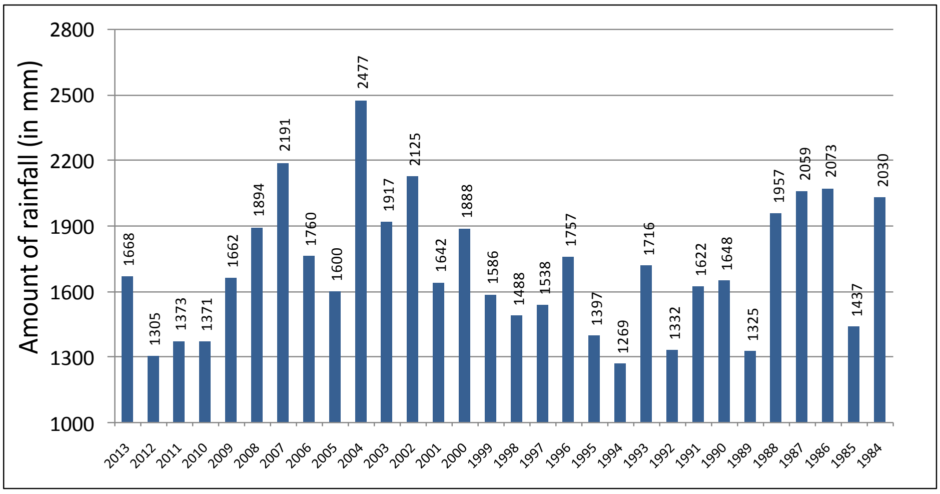

Fig-28 : Yearly total amount of rainfall for the period of 1984 to 2013

7. **Analysis**. It is revealed from this graph that the variability in rainfall amount in each year is very high. The lowest amount of rain was found 1269mm in 1994 and the highest amount was found 2477mm in 2004. A consistency was found within the years 2010 to 2012 where rainfall was 1300mm to 1400mm but in 2013, again rainfall increased and it was similar to 2009. However, **the yearly average amount of rainfall is 1703mm.**

8. **Yearly Total No of Rainy Days**. The yearly total no of rainy days found in the period of Jan 1984 to Dec 2013 has been used in this graph. The rainfall amount 1mm within 24hrs has been considered as rainy day.

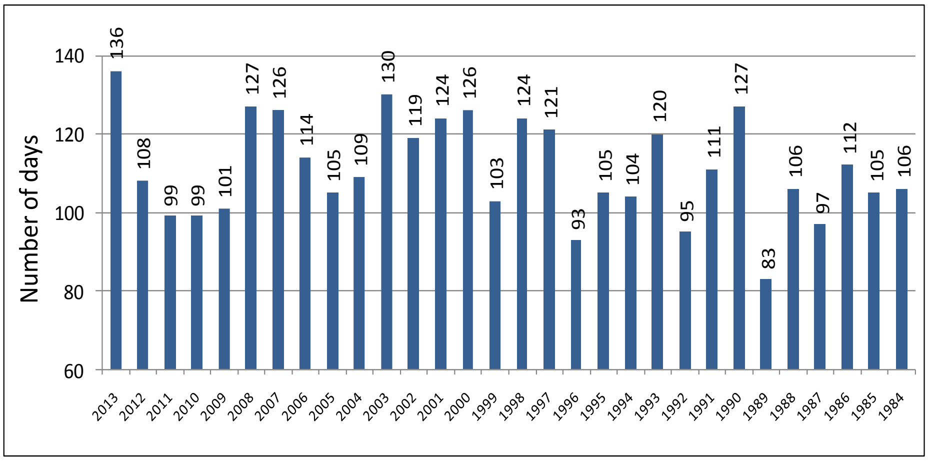

Fig-29 : Yearly total no of rainy days for the period of 1984 to 2013

9. **Analysis**. It is revealed from the study of this graph (Fig-29) that the variability in yearly total no of rainy days is also high but it is less than the variability in yearly amount of rainfall (Fig-28). The lowest no of rainy days was found 83 days in 1989 and the highest no of rainy days was found 136 days in 2013. The total no of rainy days was found within 100 days to 130 days in most of the year. However, **the average no of rainy days was found 111.1 days**.

Pentad Average Rainfall Amount and Rainy Days

10. The pentad average (five days mean) is obtained by averaging the data of each five days in a month for the period 01 Jan 1984 to 31 Dec 2013. The pentad is a world-wide used unit in dealing with meteorological phenomena, particularly in the tropics. The pentad analysis of precipitation is used for determining the onset and withdrawal dates of the rainy seasons, to represent temporal and spatial variations of precipitation in several regions over the globe, for examining the annual, inter-annual, and intra-seasonal variability of global precipitation etc. In this study, the panted average no of rainy days and rainfall amounts have been considered from April to October, as the rainfall in other months (Nov to March) are very less.

11. **Pentad Average Rainy Days for April**. The no of rainy days within each five days of April is added separately for 30 years and divided by 30 for obtaining these values.

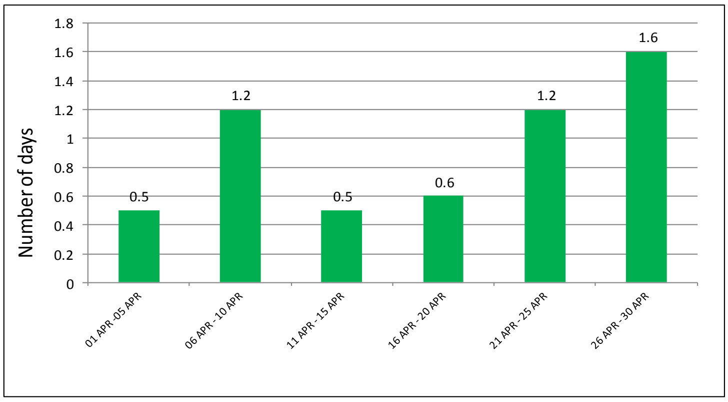

Fig-30 : Pentad average no of rainy days in April for the period of 1984 to 2013

12. **Pentad Average Rainfall of April**. The amount of rainfall within each five days of April is added separately for 30 years and divided by 30 for obtaining these values.

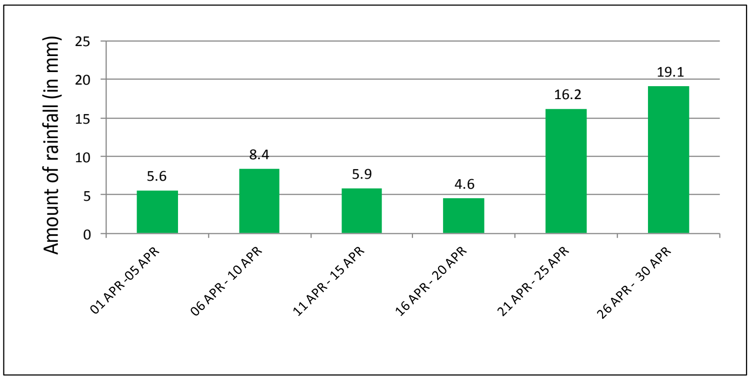

Fig-31 : Pentad average rainfall of April for the period of 1984 to 2013

13. **Analysis**. The graphs (Fig-30 & Fig-31) give the idea about the increase or decrease in no of rainy days and rainfall amount within each five days of the month. The graphs show that the 2nd pentad (06 Apr-10 Apr) has more than one day rainfall, whereas the 1st, 3rd and 4th pentad have less than one day rain; at the same time rainfall amount was also more in 2nd pentad. The rainfall increases towards the end of the month and it is more than one day. The rainfall amount also increased significantly (19.2mm) at the end period. Therefore, the 2nd, 5th and 6th pentad are vulnerable for occurrence of Nor'wester as April is pre-monsoon period.

14. **Pentad Average Rainy Days for May**. The no of rainy days within each five days of May is added separately for 30 years and divided by 30 for obtaining these values.

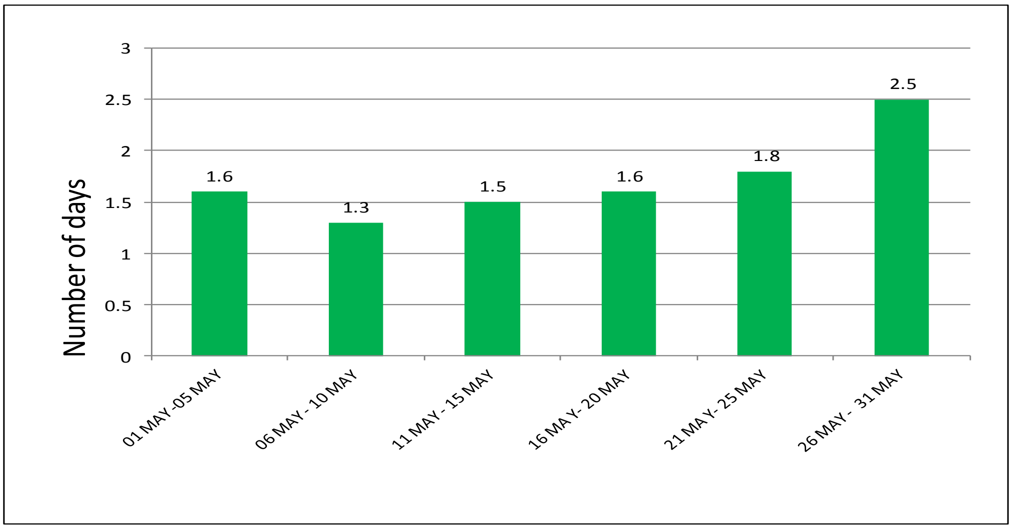

Fig-32 : Pentad average no of rainy days of May for the period of 1984 to 2013

15. **Pentad Average Rainfall of May**. The amount of rainfall within each five days of May is added separately for 30 years and divided by 30 for obtaining these values.

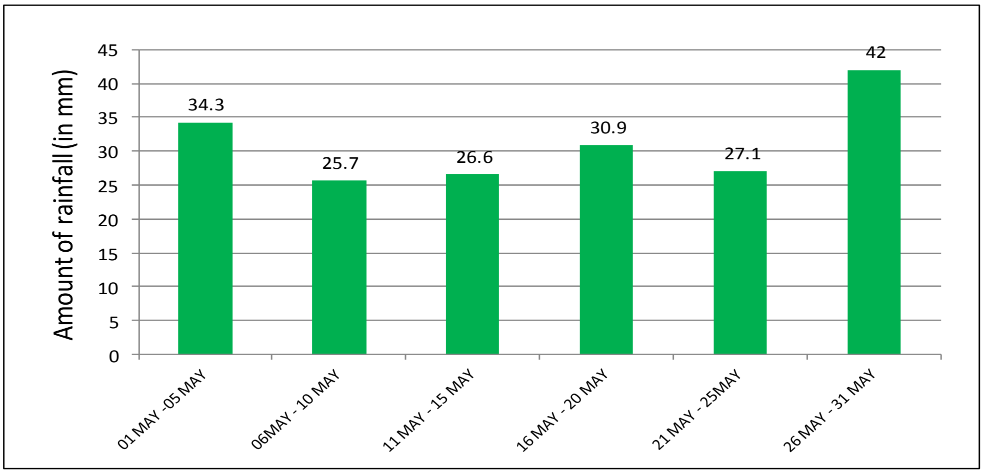

Fig-33 : Pentad average rainfall of May for the period of 1984 to 2013

16. **Analysis**. The graph (Fig-32) shows that the rainy days in all pentad was more than one day. At the beginning, rainy day was 1.6 but at the 2nd pentad it decreased to 1.3 day. Thereafter, the rainy days increases gradually and at the last pentad it increased significantly; the value was 2.5 days. The rainfall amount (Fig-33) also shows almost same characteristics of rainy days. At the last pentad (26 May- 31May), the rainfall amount was 42mm which is significantly more than other pentads of the month. The month May is the pre-monsoon period and most of the rainfall is associated with Nor'wester. Therefore, it is revealed that the 1st, 4th and 6th pentads are vulnerable for occurrence of Nor'wester.

17. **Pentad Average Rainy Days for June**. The no of rainy days within each five days of June is added separately for 30 years and divided by 30 for obtaining these values.

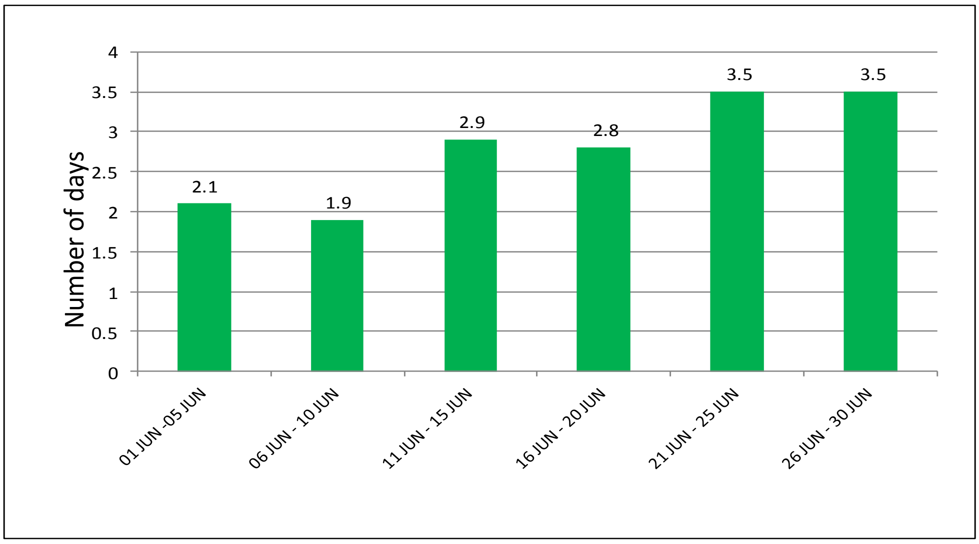

Fig-34 : Pentad average no of rainy days of June for the period of 1984 to 2013

19. **Pentad Average Rainfall of June**. The amount of rainfall within each five days of June is added separately for 30 years and divided by 30 for obtaining these values.

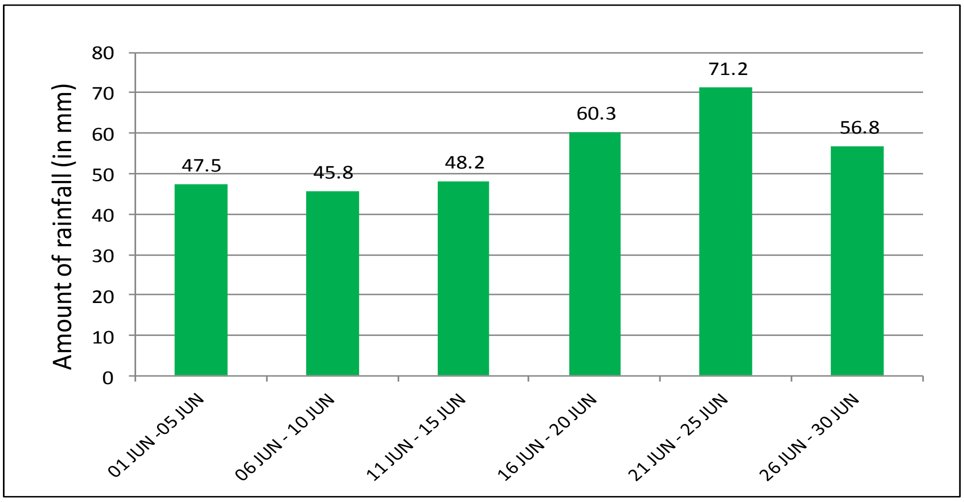

Fig-35 : Pentad average rainfall of Jun for the period of 1984 to 2013

20. **Analysis**. The graph (Fig-34) shows that the 1st and 2nd pentad have 2.1 and 1.9 day rain; thereafter the rainy days has increased gradually up to 3.5 days at the end. The graph (Fig-35) shows that rainfall amount in the 1st, 2nd and 3rd pentad are almost similar, thereafter increased gradually. The highest amount of rain was found in the 5th pentad (21 Jun - 25 Jun) 71.2mm. At the last pentad, rainfall amount has decreased up to 56.8mm. The 5th pentad (21 Jun – 25 Jun) and 6th pentad (26 Jun – 30 Jun) have 3.5 days rain each and amount of rainfall 71.2mm & 56.8mm respectively. Therefore, these two pentads are susceptible for mostly cloudy and continuous rainy days.

21. **Pentad Average Rainy Days for July**. The no of rainy days within each five days of July is added separately for 30 years and divided by 30 for obtaining these values.

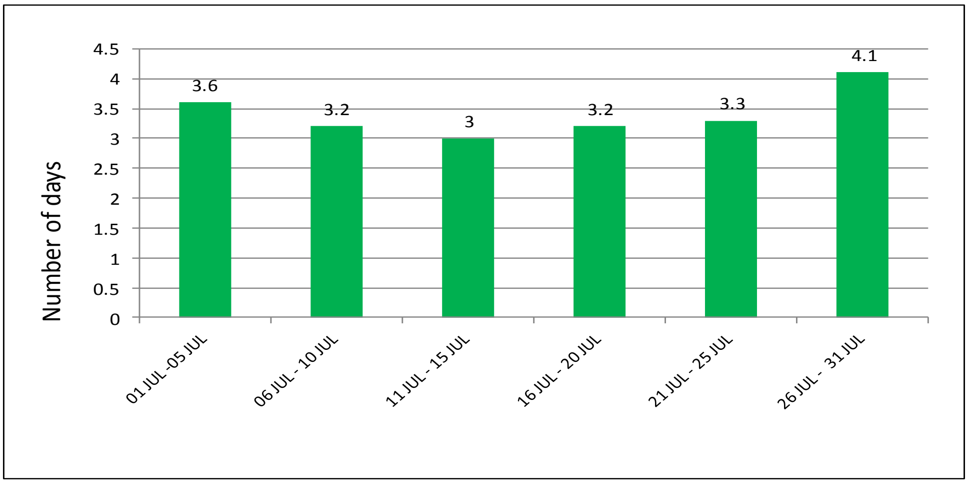

Fig-36 : Pentad average no of rainy days of July for the period of 1994 to 2013

22. **Pentad Average Amount of Rainfall for July**. The amount of rainfall within each five days of July is added separately for 30 years and divided by 30 for obtaining these values.

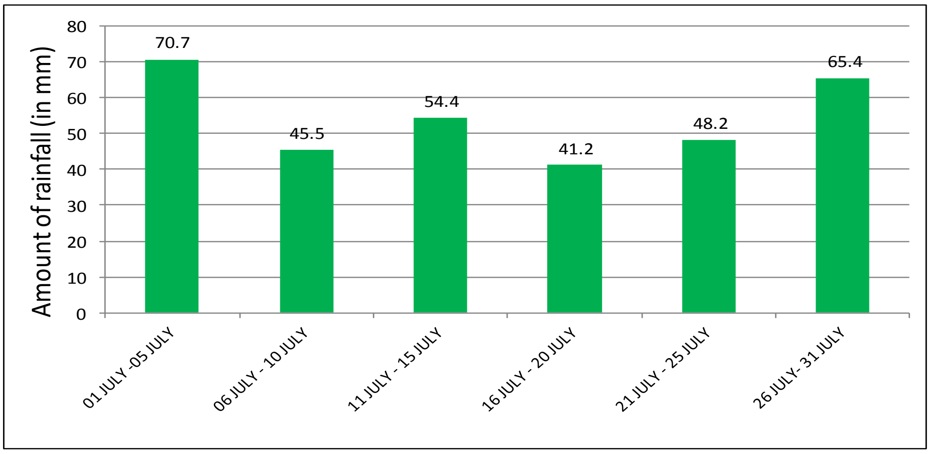

Fig-37 : Pentad average rainfall of July for the period of 1984 to 2013

23. **Analysis**. The graph (Fig-36) shows that, throughout the month rain was 3 days or more in each pentad. The 3rd pentad has 3 days rain, which is less in comparison with other pentads of the month. At the 1st and last pentad, rainy days were 3.6 and 4.1 respectively. The next graph (Fig-37) shows that the highest amount of rain (70.7mm) was in the 1st pentad and the second highest rainfall (65.4mm) was in the last pentad. Therefore, the 1st and last pentads are susceptible for continuous rainy days when human daily activities are hampered. The minimum rainfall in the month (41.2mm) was in the 4th pentad. Therefore, from these two graphs (Fig-36 & Fig-37), it reveals that the 2nd, 3rd and 4th pentad are susceptible for one or two break in monsoon.

24. **<u>Pentad Average Rainy Days for August</u>**. The no of rainy days within each five days of August is added separately for 30 years and divided by 30 for obtaining these values.

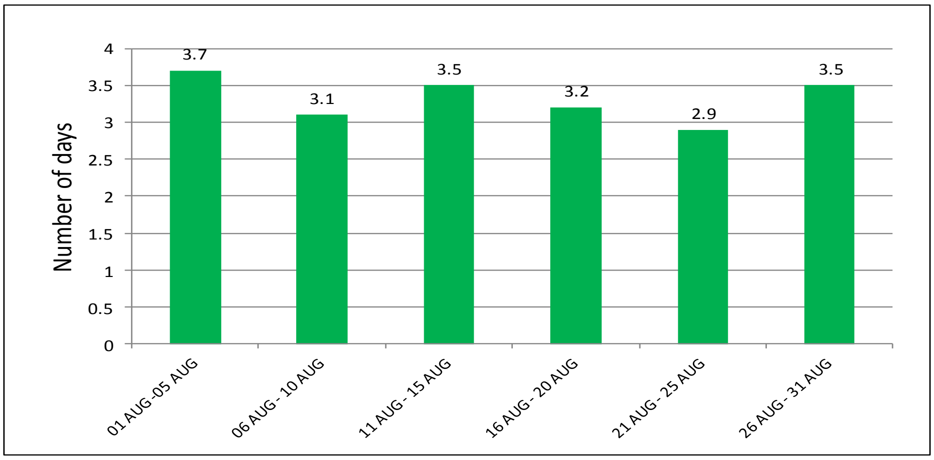

Fig-38 : Pentad average no of rainy days of August for the period of 1984 to 2013

25. **<u>Pentad Average Amount of Rainfall for August</u>**. The amount of rainfall within each five days of August is added separately for 30 years and divided by 30 for obtaining these values.

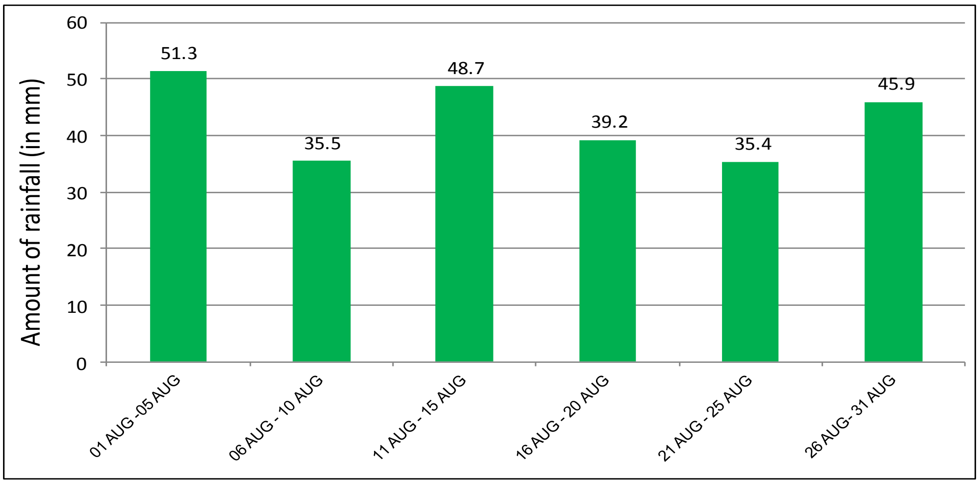

Fig-39 : Pentad average rainfall of August for the period of 1984 to 2013

26. **<u>Analysis</u>**. The graph (Fig-38) shows that the 1st pentad (01 Aug – 05 Aug) has maximum rainy days (3.7) and in the 2nd pentad it has reduced to 3.1 days. In the 3rd and 4th pentad it became 3.5 and 3.2 days respectively, but in 5th pentad it reduced to 2.9 days. Again, it increased to 3.5 days at the end. The next graph (Fig-39) shows that the maximum amount of rainfall (51.3mm) was in the 1st pentad (01 Aug – 05 Aug) and in the 2nd pentad it reduced to 35.5mm. Again it increased in 3rd pentad (48.7mm) and reduced in 5th pentad (35.4mm). Therefore, it is revealed from the graphs that the 2nd (06 Aug – 10 Aug) and 5th (21 Aug – 25 Aug) pentads are susceptible for break monsoon. It is also revealed that the 1st, 3rd and last pentads are susceptible for continuous rainy days with mostly cloudy sky.

27. **Pentad Average Rainy Days for September**. The no of rainy days within each five days of September is added separately for 30 years and divided by 30 for obtaining these values.

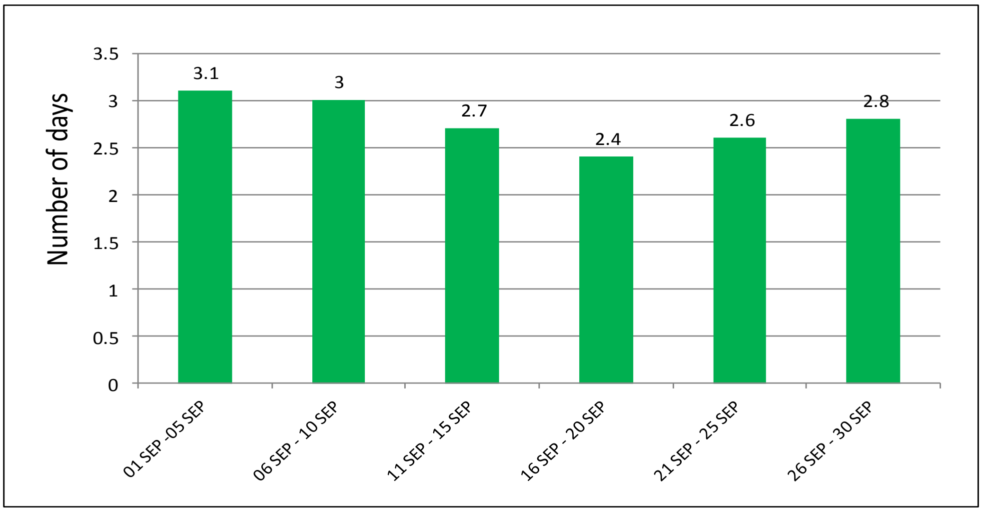

Fig-40 : Pentad average no of rainy days of September for the period of 1984 to 2013

28. **Pentad Average Rainfall for September** The amount of rainfall within each five days of September is added separately for 30 years and divided by 30 for obtaining these values.

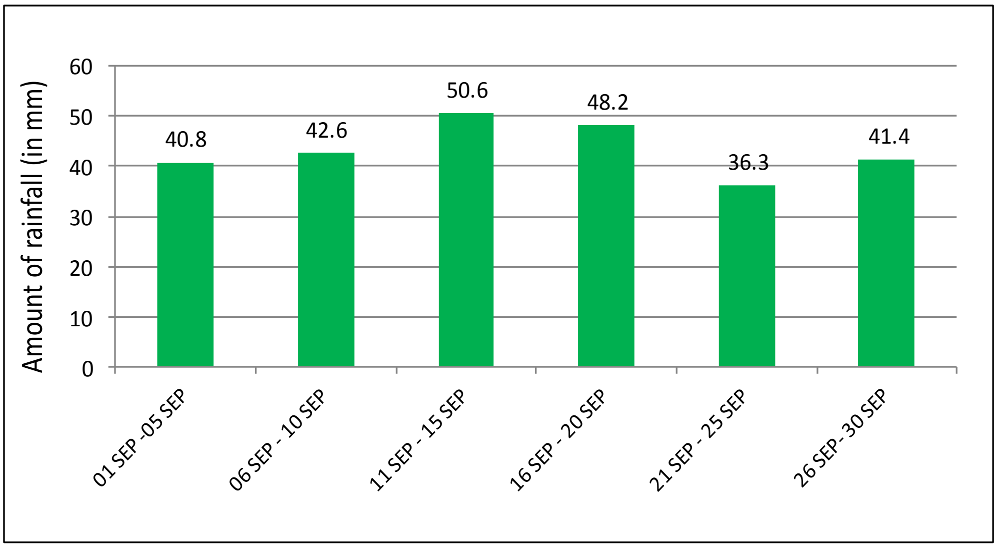

Fig-41 : Pentad average rainfall of September for the period of 1984 to 2013

29. **Analysis**. The graph (Fig-40) shows that the maximum no of rainy days (3.1) was in the 1st pentad (01 Sep – 05 Sep) and gradually the no of days decreased to 2.4 days in the 4th pentad (16 Sep – 20 Sep). Again the days increased to 2.8 days at the end. The next graph (Fig-41) shows that the 1st pentad (01 Sep – 05 Sep) has 40.8mm rainfall which increased gradually and maximum rainfall (50.6mm) was in 3rd pentad (11 Sep – 15 Sep). Thereafter, the amount of rainfall decreased and minimum rainfall (36.3mm) was in the 5th pentad (21 Sep – 25 Sep). Therefore, it is revealed from the graphs (Fig-40 & Fig-41) that the 5th pentad (21 Sep – 25 Sep) is susceptible for break in monsoon.

30. **<u>Pentad Average Rainy Days for October</u>**. The no of rainy days within each five days of October is added separately for 30 years and divided by 30 for obtaining these values.

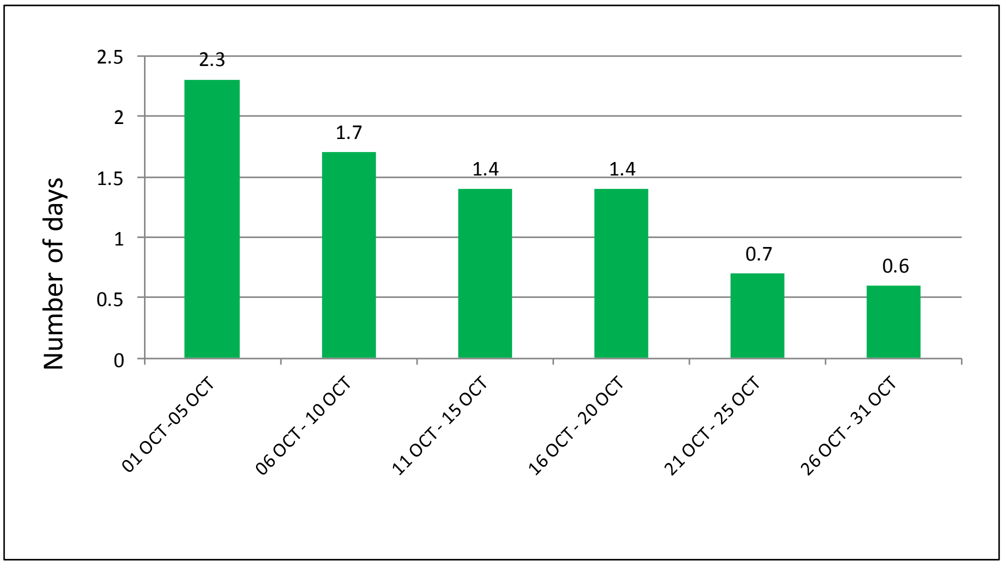

Fig-42 : Pentad average no of rainy days of October for the period of 1984 to 2013

31. **<u>Pentad Average Rainfall of October</u>**. The amount of rainfall within each five days of October is added separately for 30 years and divided by 30 for obtaining these values.

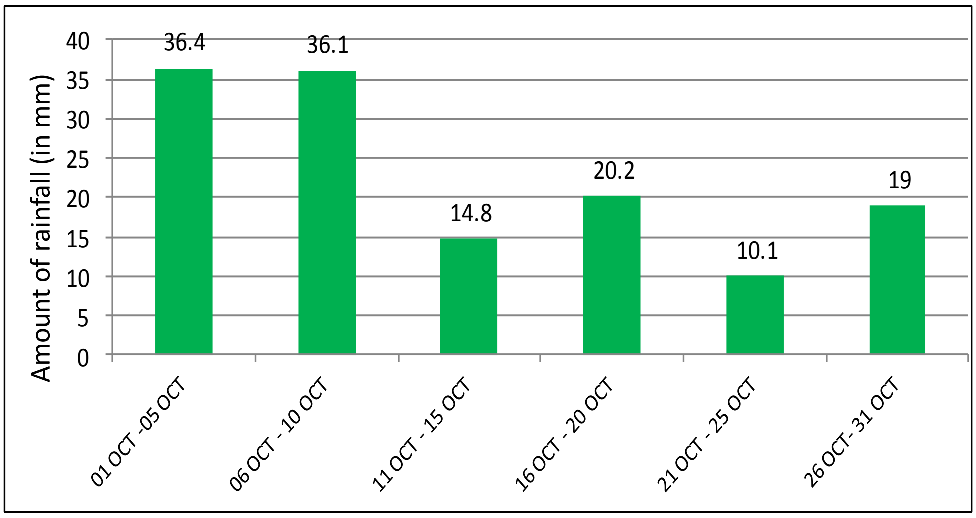

Fig-43 : Pentad average rainfall of October for the period of 1984 to 2013

32. **<u>Analysis</u>**. The graph (Fig-42) shows that the 1st pentad (01 Oct – 05 Oct) has maximum no of rainy days (2.3 days) and thereafter it decreases gradually up to 0.6 days at the end period of the month. The next graph Fig-43) shows that the 1st and 2nd pentad have 36.4mm and 36.1mm rainfall respectively, thereafter the rainfall decreased significantly which indicates the withdrawal of monsoon.

Daily Average Amount of Rainfall and Rainy Days

33. **Daily Average Amount of Rainfall in February**. Daily Average amount of rainfall has been calculated by averaging the daily amount of rainfall for last 30 years. As the month is under pre-monsoon period, it is considered that almost all of the rainfall is associated with Nor'wester.

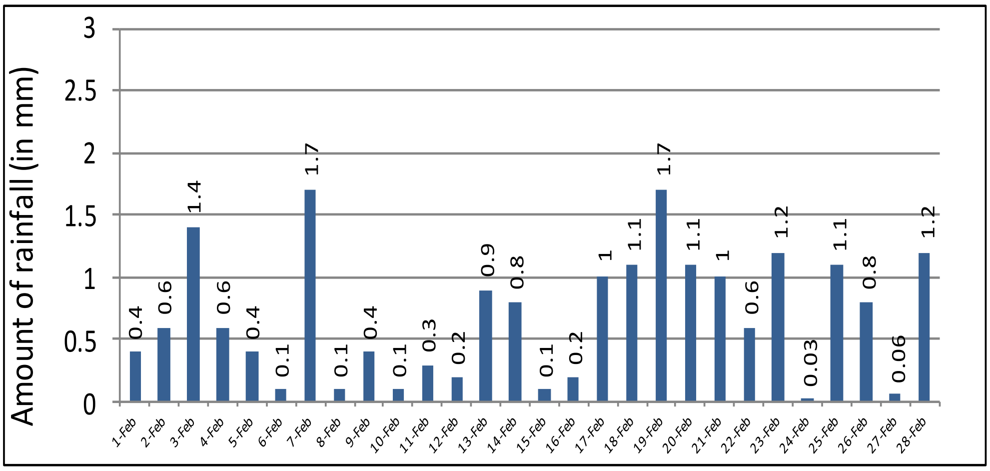

Fig-44 : Daily average amount of rainfall for the period of 1984 to 2013

34. **No of Rainy Days in February**. It is the date-wise total no of rainy days within last 30 years.

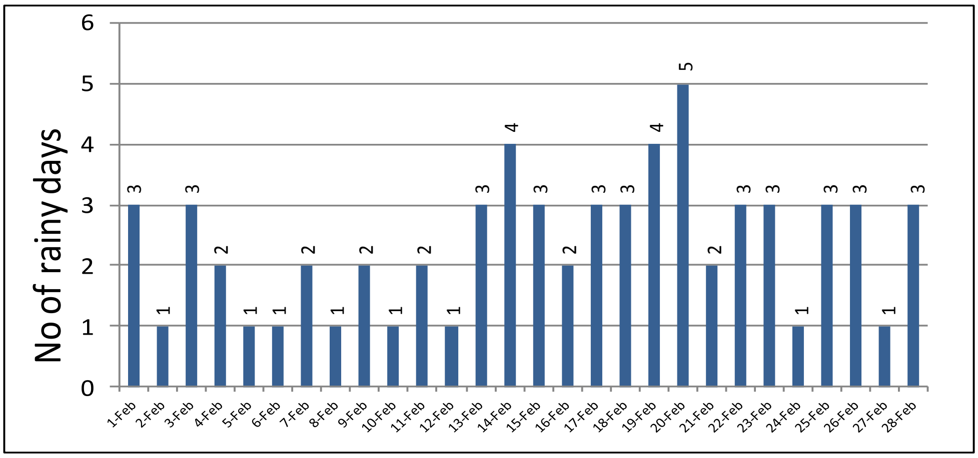

Fig- 45 : Total No of rainy days for the period of 1984 to 2013

35. **Analysis**. February is the per-monsoon period and most of the rainfall is associated with nor'wester. The frequency of Nor'wester in February is very less (0.5). It is observed from the data that the years when one Nor'wester occurred early in February, in those years, series of Nor'westers occurred in the rest of the month. Total 12 years Nor'wester occurred in varying numbers out of 30 years but rainfall occurred in 20 years. It is revealed from the graphs (Fig-44 & Fig-45) that the 19 and 20 February are vulnerable for occurrence of Nor'wester considering the no of rainy days and amount of Rainfall. From the analysis of "nor'wester" in "Thunderstorm Behaviour" chapter, it was found that the 19th, 25th and 28th February had some probability (<10%) of nor'wester. Combination of these two, it can be considered that 19th, 20th, 25th and 28th February are the susceptible days for nor'wester. (Probability 10% -15%)

36. **<u>Daily Average Amount of Rainfall in March</u>**. Daily Average rainfall amount has been calculated by averaging the daily amount of rainfall for 30 years. The month March is under pre-monsoon season; therefore, it is considered that the rainfall is associated with Nor'wester.

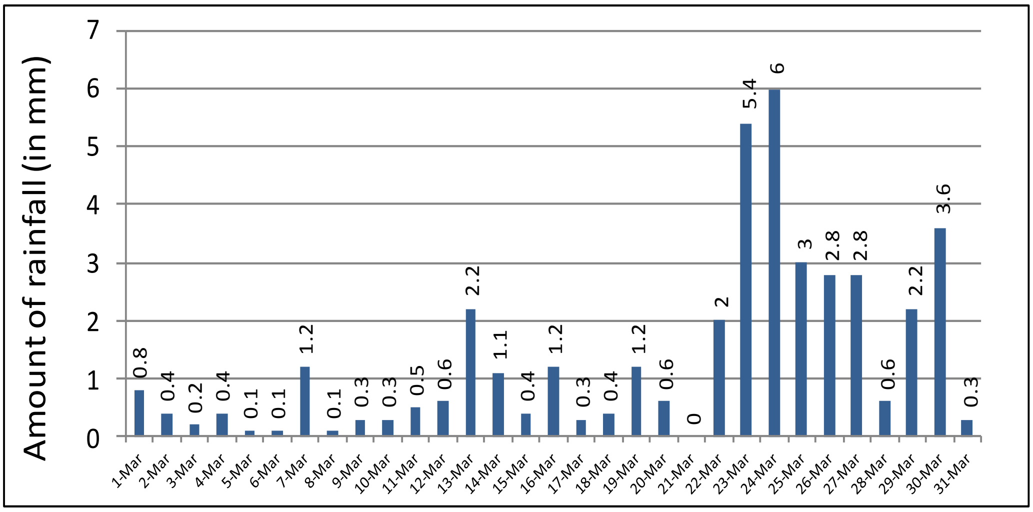

Fig-46 : Daily average amount of rainfall for the period of 1984 to 2013

37. **<u>No of Rainy Days in March</u>**. It is the date-wise total no of rainy days within last 30 years.

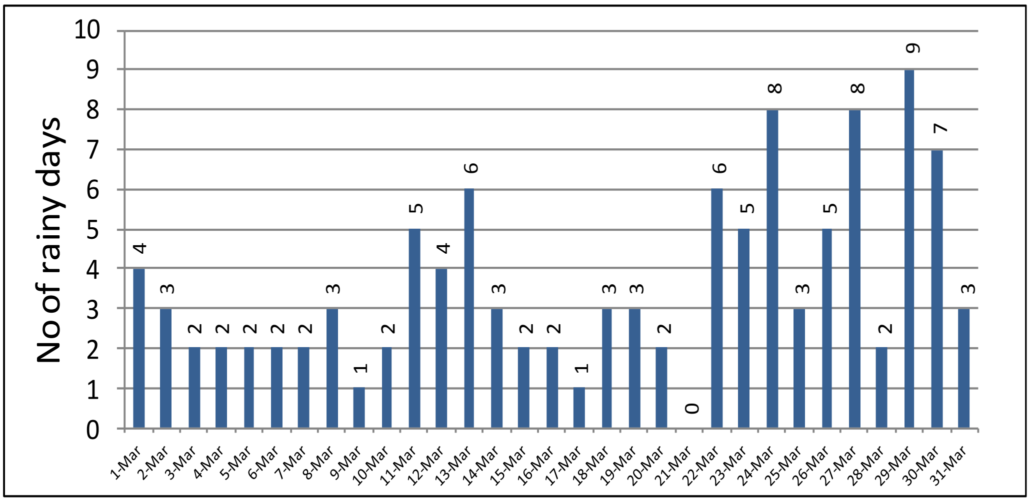

Fig-47 : Total No of rainy days for the period of 1984 to 2013

38. **<u>Analysis</u>**. The average frequency of Nor'wester in March is 2.2 (Fig-18). Total 23 years Nor'wester occurred in varying numbers out of 30 years. It is revealed from the study of the graphs (Fig-46 & Fig-47) that the average amount of rainfall and rainy days have increased significantly from 22 March. Therefore, the Nor'wester mainly starts from 22 March, though few have occurred early also. The susceptible dates for occurrence of Nor'wester can be considered as 23rd, 24th, 27th, 29th and 30th March, after considering the average rainfall amount and no of total rainy days. From the analysis of "nor'wester" in "Thunderstorm Behaviour" chapter, the susceptible days were found 29th and 30th March, which commensurate with this analysis. Combination of these two, the susceptible days can be considered as 23rd, 24th, 27th, 29th and 30th March (probability 20% - 30%).

39. **Daily Average Amount of Rainfall in April**. Daily Average rainfall amount has been calculated by averaging the daily amount of rainfall for 30 years. As April is the pre-monsoon period, therefore, almost all rainfall is associated with Nor'wester and the graphs (Fig-48 & Fig-49) give the idea about the occurrence of Nor'wester.

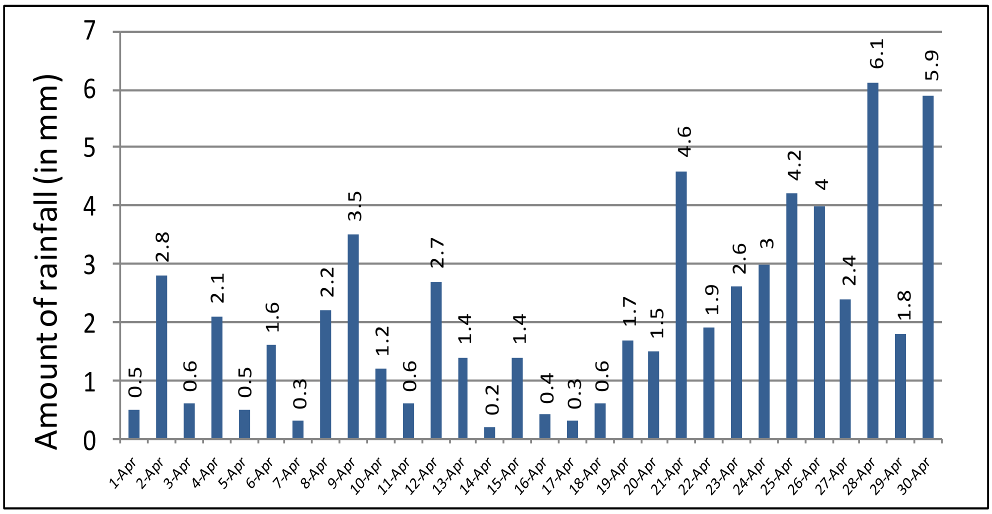

Fig- 48 : Daily average amount of rainfall for the period of 1984 to 2013

40. **No of Rainy Days in April**. It is the date-wise total no of rainy days within last 30 years.

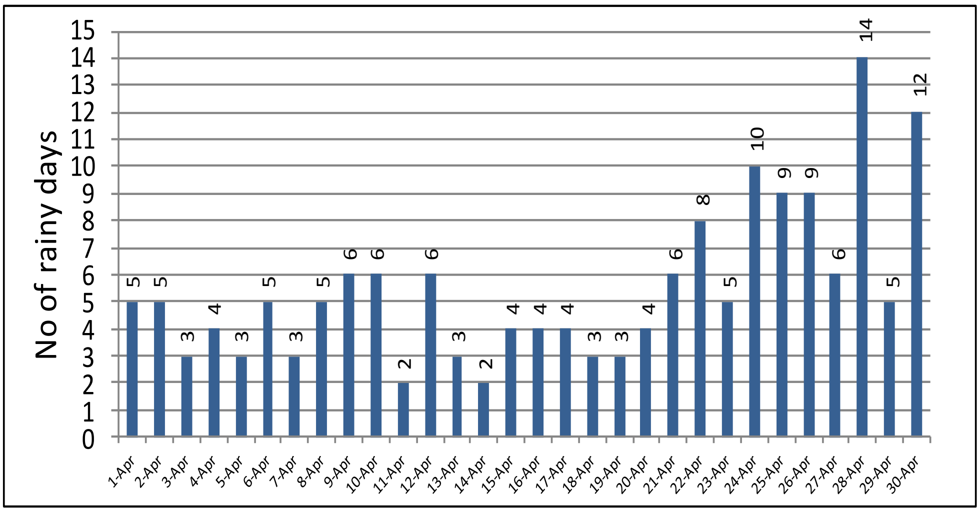

Fig- 49 : Total no of rainy days for the period of 1984 to 2013

41. **Analysis**. The average frequency of Nor'wester in April is 5.5 (Fig-18). Though there is no hard and fast rule for finding out the Nor'wester days, but the amount of rainfall against the no of rainy days gives an idea that the 09th, 12th, 21st, 24th, 25th, 26th, 28th and 30th April are vulnerable for occurrence of Nor'wester. From the analysis of "nor'wester" in "Thunderstorm Behaviour" chapter, the susceptible days were found 22nd, 23rd, 26th, 28th and 29th April and the probability is around 30%. Now, from the combination of these two, the susceptible days of nor'wester can be considered as 22nd, 24th, 26th, 28th, 29th and 30th April and the probability is around 30%.

42. **Daily Average Amount of Rainfall for May**. Daily Average rainfall amount of May has been calculated by averaging the daily amount of rainfall for 30 years. The month May is under pre-monsoon season and the rainfall is mostly from Nor'wester.

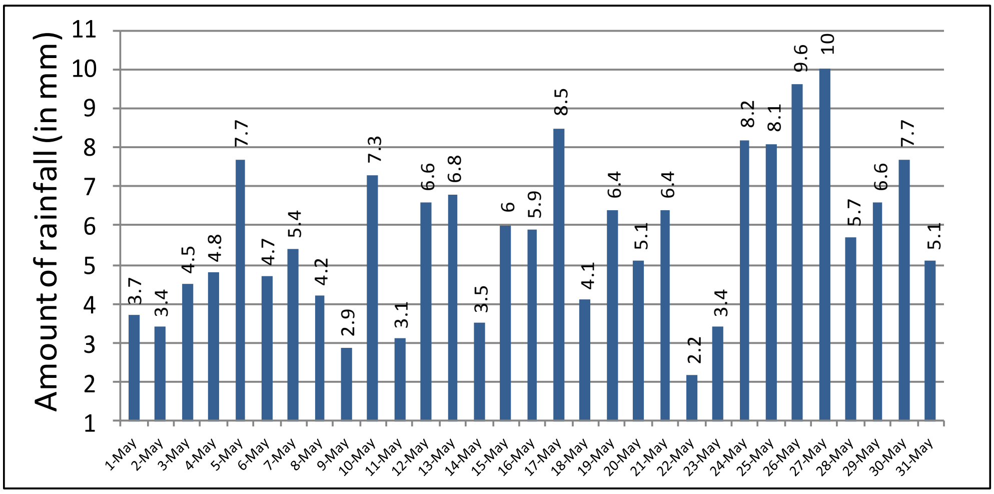

Fig- 50 : Daily average amount of rainfall for the period of 1984 to 2013

43. **No of Rainy Days in May**. The date-wise total no of rainy days within last 30 years have been used directly in this graph.

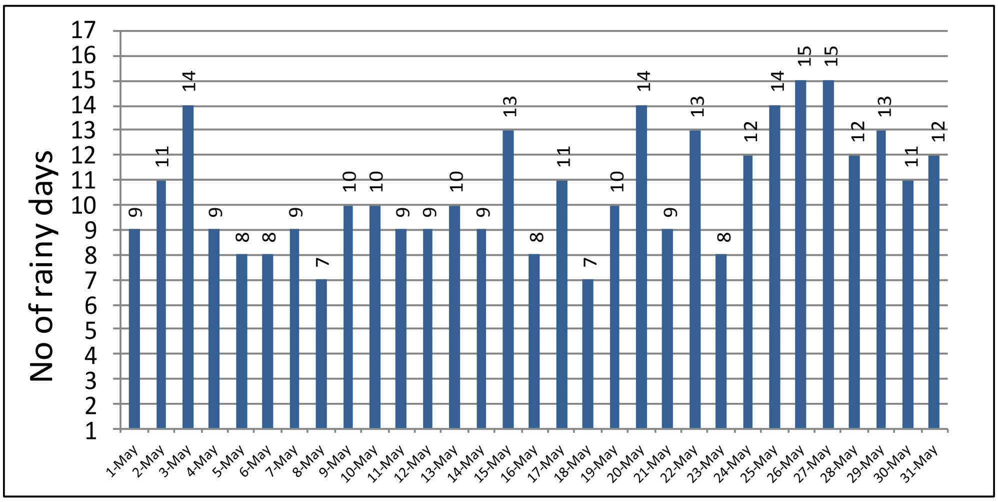

Fig-51 : Total no of rainy days forthe period of 1984 to 2013

44. **Analysis**. The frequency of Norwester is highest (average 8.5) in the month of May. The graphs (Fig-50 & Fig-51) show the average amount of rainfall and average no of rainy days respectively within last 30 years (1984 to 2013). The rainy days all are not Nor'wester days. Considering the no of rainy days and average amount of rainfall, it is revealed from the study of the graphs that the 12th, 13th, 17th, 20th, 24th, 25th, 26th, 27th, 28th and 29th May are vulnerable for occurrence of Nor'wester. From the analysis of "nor'wester" in "Thunderstorm Behaviour" chapter, the susceptible days were found 2nd, 3rd, 5th, 7th, 8th, 10th, 13th, 15th and 24th May and the probability was ≥30%. Now, from the combination of these two, the susceptible days of nor'wester can be considered as 3rd, 5th, 10th, 13th, 15th, 17th, 24th, 25th, 26th and 27th April (probability is 30% - 40%).

45. **Daily Average Amount of Rainfall for June**. Daily Average rainfall amount of June has been calculated by averaging the daily amount of rainfall for 30 years.

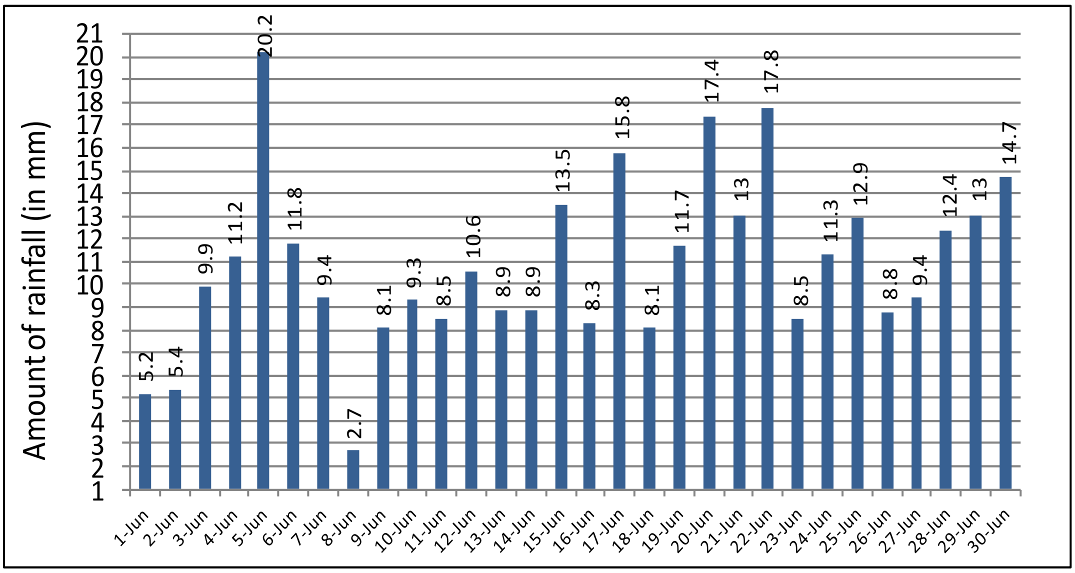

Fig-52 : Daily average amount of rainfall for the period of 1984 to 2013

46. **No of Rainy Days in June**. The day-wise total no of rainy days of June within last 30 years have been used directly in this graph.

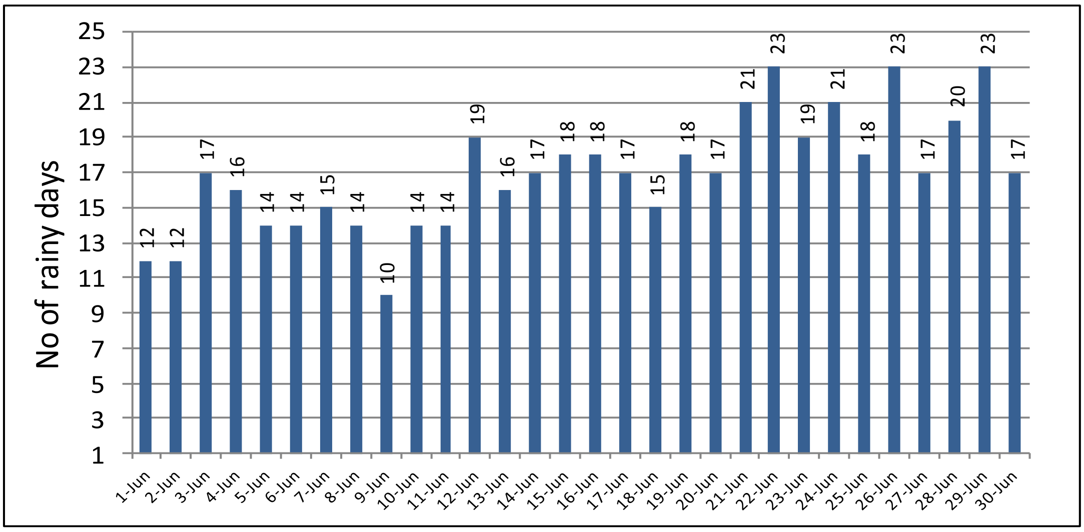

Fig-53 : Total no of rainy days for the period of 1984 to 2013

47. **Analysis**. Nor'wester occurs in the 1st and 2nd week of June also till monsoon is set over the country. After onset of monsoon, rainy days increases significantly. The graphs (Fig-52 & Fig-53) show the significant change of rainy days from 12 June onward and it may be the onset date of monsoon over Jessore area. After onset of monsoon, the 17th, 20th, 22nd, 25th and 29th June are susceptible for continuous rainy days, when normal activities may be hampered due to rain. The graphs also reveal that the 04th, 05th, 07th, 10th and 11th June are vulnerable for occurrence of Nor'wester before onset of monsoon. On 05th Jun, average rainfall amount is very high (20.2mm); because, an unusual amount of rainfall (242mm) occurred on 05 Jun 1985.

48. **Daily Average Amount of Rainfall for July**. Daily Average rainfall amount of July has been calculated by averaging the daily amount of rainfall for 30 years and used in the graph.

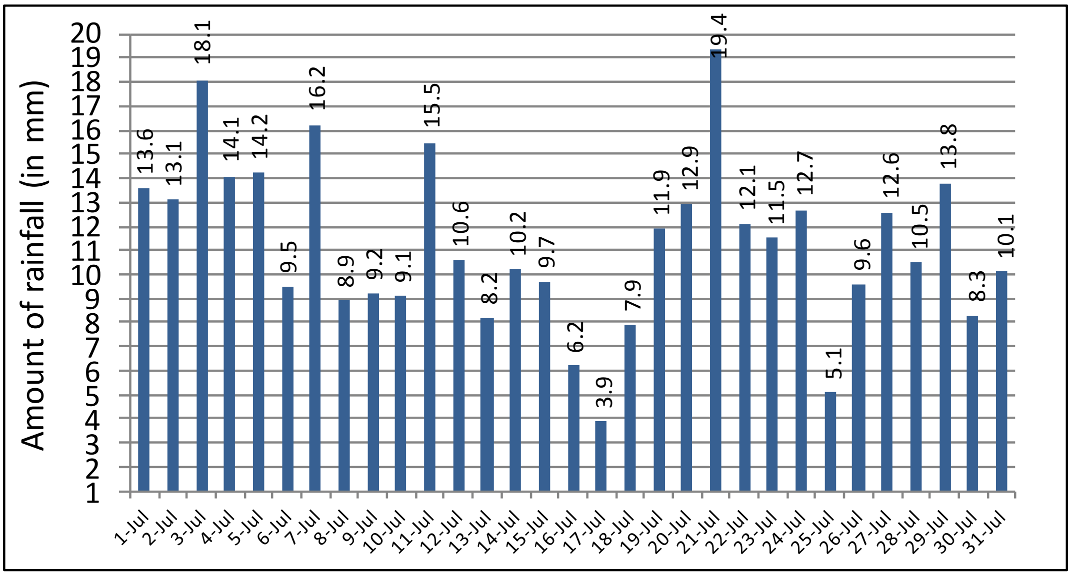

Fig-54 : Daily average amount of rainfall for the period of 1984 to 2013

49. **No of Rainy Days in July**. The day-wise total no of rainy days of July within last 30 years have been used directly in the graph.

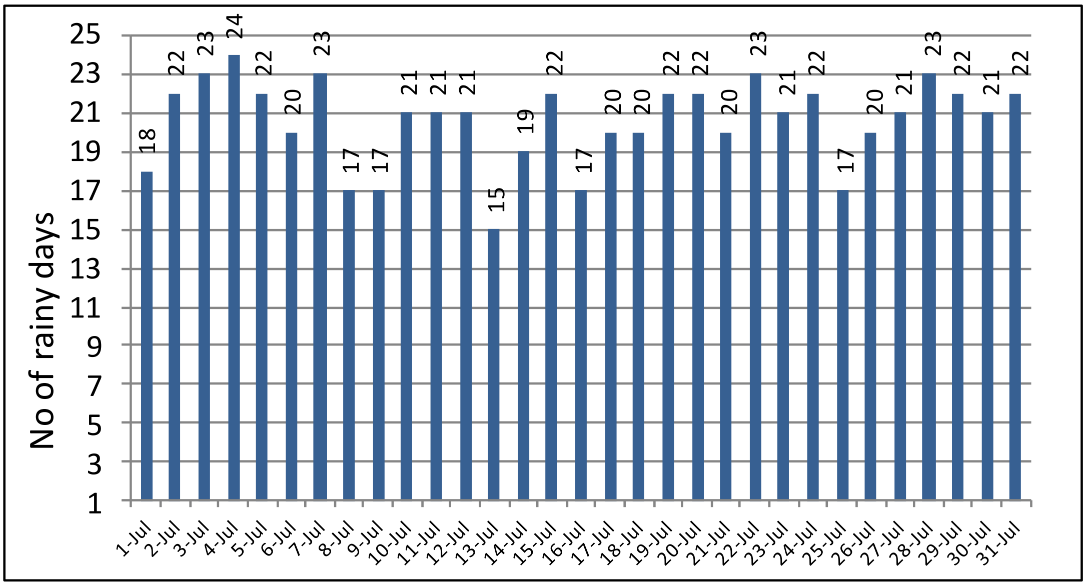

Fig-55 : Total no of rainy days for the period of 1984 to 2013

50. **Analysis**. The month July has the highest average no of rainy days (22 days). The graphs (Fig-54 & Fig-55) reveal that the 04, 15, 19, 21, 23, 28 and 29 July are susceptible for continuous rainy days with overcast sky when normal activities are hampered. Here, the days having rainfall ≥5mm for ≥16 days out of 30 days have considered susceptible for continuous rain. It also reveals that the 08 Jul - 09 Jul are susceptible for one break and 16 Jul - 17 Jul are susceptible for another break in monsoon.

51. **Daily Average Amount of Rainfall for August**. The daily average amounts of rainfall are obtained by averaging the daily amount of rainfall for last 30 years and are used in the graph.

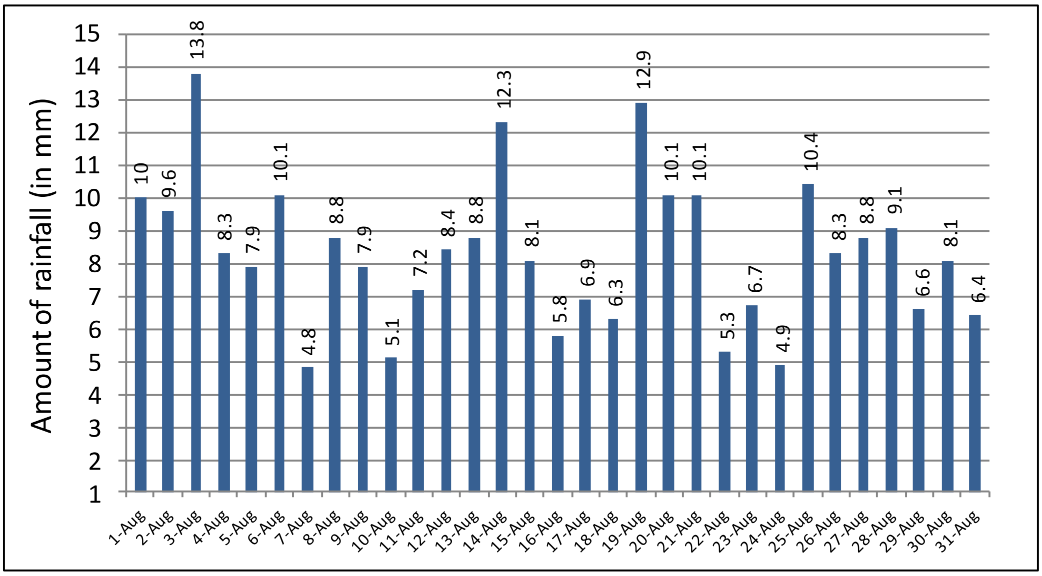

Fig-56 : Daily average amount of rainfall for the period of 1984 to 2013

52. **No of Rainy Days in August**. The day-wise total no of rainy days in August within last 30 years have been used directly in the graph.

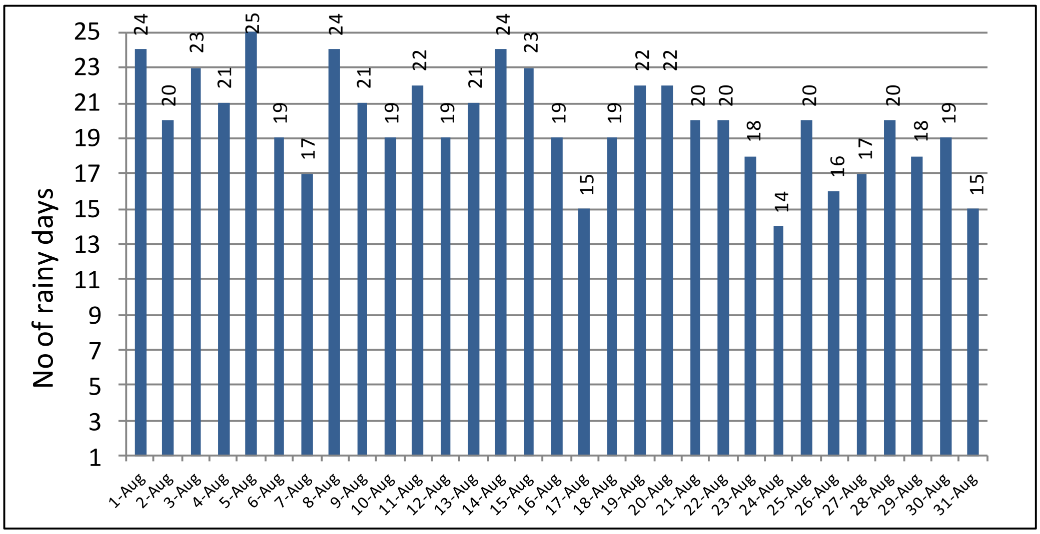

Fig- 57 : Total no of rainy days for the period of 1984 to 2013

53. **Analysis**. The average no of rainy days in August is 20 days. It is revealed from the graphs (Fig-56 & Fig-57) that 06 Aug – 07 Aug, 16 Aug – 18 Aug and 23 Aug – 24 Aug are susceptible for break in monsoon. Here, the periods having ≤10 days rainfall out 30 days have considered susceptible for break. It is also revealed that the 02, 14, 15, 19 and 28 August are susceptible for continuous rainy days with cloudy to overcast sky. Here, the days having rainfall ≥5mm for ≥16 days out of 30 days have considered susceptible for continuous rain. The continuous rainy days stop all outdoor activities including flying.

54. **Daily Average Amount of Rainfall for September**. The daily average amounts of rainfall are obtained by averaging the daily rainfall of September for last 30 years and are used in the graph.

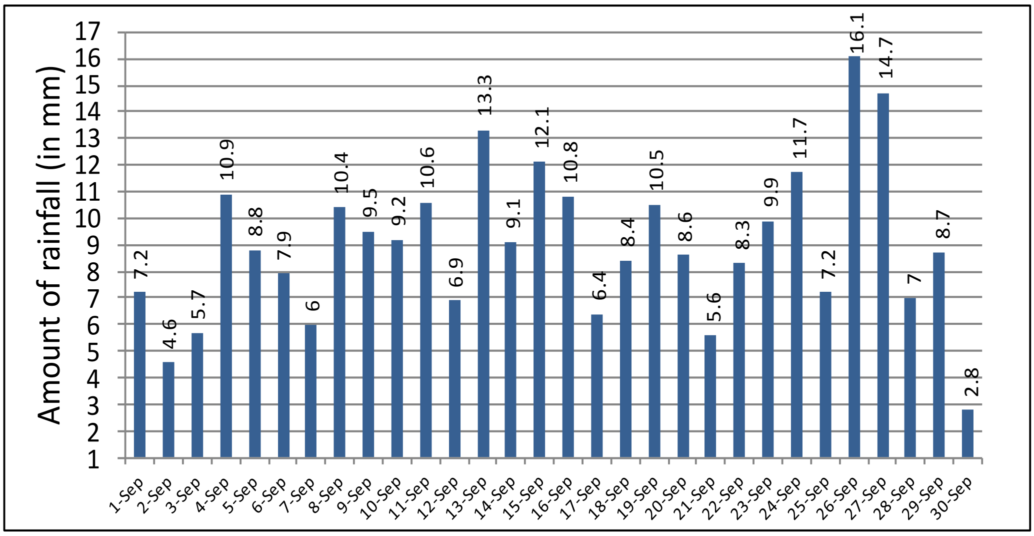

Fig-58 : Daily average amount of rainfall for the period of 1984 to 2013

55. **No of Rainy Days for September**. The day-wise total no of rainy days in September within last 30 years have been used directly in the graph.

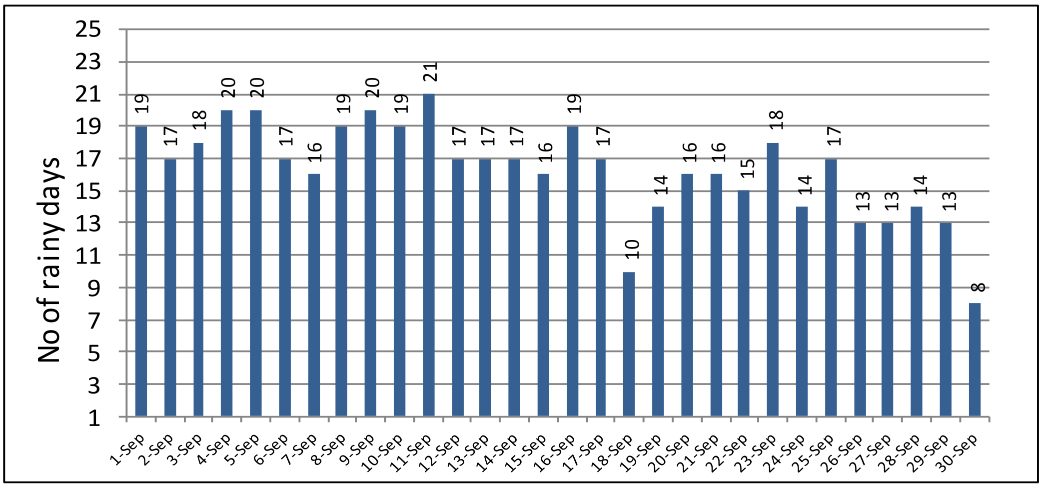

Fig-59 : Total no of rainy days for the period of 1984 to 2013

56. **Analysis**. The graphs (Fig-58 & Fig-59) reveal that the 01 Sep – 02 Sep, 06 Sep – 07 Sep, 17 Sep – 21 Sep and 28 Sep – 30 Sep are susceptible for break monsoon. Here, the periods having ≤10 days rainfall out 30 days have considered susceptible for break. On the other hand, the 05, 09 and 11 September are susceptible for continuous rainy days. Here, the days having rainfall ≥5mm for ≥14 days out of 30 days have considered susceptible for continuous rain. On 13 and 26 September, average rainfall amount is high (13.3mm and 16.1mm respectively); because, unusual amount of rainfall (215mm and 255mm) occurred on 13 Sep 04 and 26 Sep 86. During the break monsoon periods, rainfall ceases and cloud decreases. As a result, these periods could be utilized for compensating the loss occurred in the days when rainfall was continuous.

57. **Daily Average Amount of Rainfall for October**. The daily average amounts of rainfall are obtained by averaging the daily rainfall of October for last 30 years and are used in the graph.

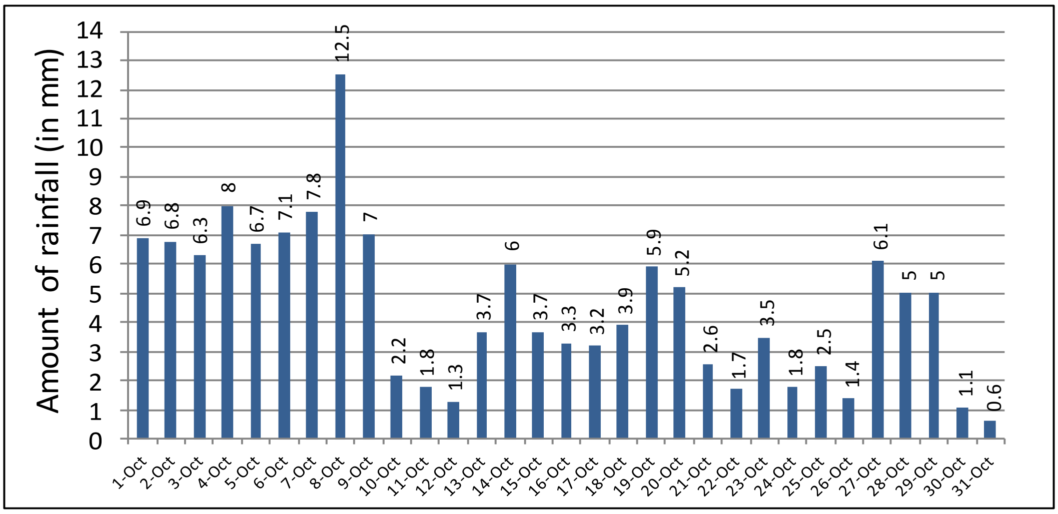

Fig-60 : Daily average amount of rainfall for the period of 1984 to 2013

58. **No of Rainy Days in October**. The day-wise total no of rainy days in October within last 30 years have been used directly in the graph.

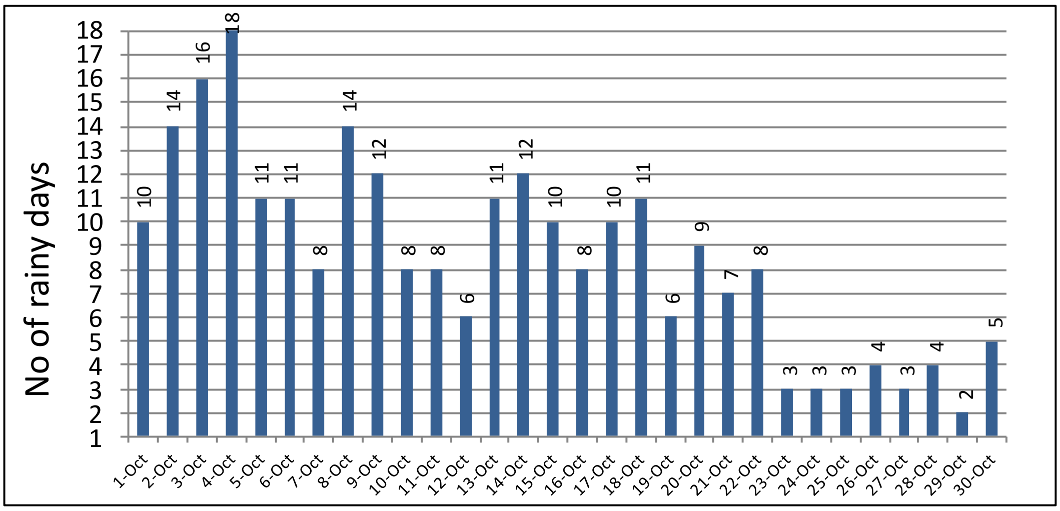

Fig-61 : Total no of rainy days for the period of 1984 to 2013

59. **Analysis**. The graphs (Fig-60 & Fig-61) shows that the rainfall amount and rainy days has reduced significantly on 10th October onward and it may be due to the withdrawal of monsoon. Therefore, 10th October may be the withdrawal date of monsoon over Jessore area.

Tables for Monthly Max Wind, Total Amount of Rainfall and Rainy Days

60. **Monthly Maximum Wind Direction & Speed (In kts)**. Monthly maximum Wind Direction & Speed (In Kts) are shown in Table-1:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1984	360/13	340/18	220/22	200/50	090/50	220/32	130/34	090/25	360/26	090/23	020/18	220/18
1985	340/10	340/10	180/30	160/50	340/35	290/27	-	-	090/17	240/26	040/10	340/12
1986	360/10	360/12	220/26	230/46	340/36	310/32	200/26	090/24	090/32	230/30	040/30	360/12
1987	020/13	340/20	230/33	180/33	360/41	180/21	230/22	020/32	200/22	070/19	050/13	180/18
1988	020/12	250/18	310/42	270/36	200/34	-	160/27	050/24	230/27	160/22	050/59	360/11
1989	360/12	230/22	310/28	220/30	160/33	040/33	230/25	150/22	120/23	120/23	360/10	050/14
1990	200/16	360/21	200/44	030/40	020/42	050/32	110/24	090/29	180/24	080/20	310/28	080/27
1991	360/13	360/33	020/37	340/36	350/44	350/30	270/26	130/33	210/24	110/20	110/10	020/18
1992	340/18	350/28	280/20	230/35	240/33	160/26	180/30	100/24	180/22	050/18	010/10	320/08
1993	180/12	130/21	250/34	340/41	340/40	350/25	310/25	190/20	200/24	190/20	180/12	020/08
1994	360/20	020/20	200/31	360/52	360/42	360/32	150/20	110/27	040/25	180/19	050/12	360/09
1995	030/17	230/18	270/27	290/30	130/24	030/40	130/21	090/20	140/24	130/24	090/21	330/20
1996	360/12	310/30	320/20	320/32	320/37	190/30	180/21	030/18	360/20	060/23	130/23	330/09
1997	270/11	340/17	270/36	230/39	330/36	330/32	240/21	120/30	350/20	270/09	030/12	340/13
1998	350/10	340/20	270/28	360/46	360/33	330/25	140/21	180/13	240/24	160/21	310/27	360/07
1999	350/08	360/20	220/18	060/36	190/36	100/25	250/20	080/21	120/25	130/20	160/18	030/09
2000	090/18	010/21	290/25	290/45	260/42	120/23	190/19	190/28	190/30	020/35	360/08	010/08
2001	280/11	280/30	260/37	270/42	350/40	240/32	090/22	240/32	090/29	140/26	090/12	360/08
2002	220/09	320/22	300/24	010/35	360/40	340/40	330/30	090/20	130/20	360/20	090/25	360/10
2003	360/12	360/35	360/35	360/38	240/43	360/39	100/30	100/21	130/18	190/21	350/07	350/09
2004	350/19	200/12	250/30	330/40	330/44	050/32	220/23	100/20	170/28	150/20	360/10	360/10
2005	090/12	300/30	220/50	270/35	040/30	230/28	280/23	100/20	170/21	120/20	360/11	360/10
2006	340/11	210/13	230/18	350/40	350/30	220/18	110/21	030/30	270/26	120/17	030/08	360/07
2007	350/16	030/24	030/22	250/38	360/35	020/28	160/19	150/19	120/22	240/18	030/35	350/07
2008	340/13	060/26	270/40	290/30	350/48	190/30	270/22	200/21	080/20	020/30	010/09	030/07
2009	230/10	350/16	010/31	190/22	080/48	270/22	130/20	140/18	150/30	180/10	040/06	340/10
2010	020/16	290/26	360/37	010/35	350/45	350/40	100/18	090/18	150/18	150/26	350/16	350/10
2011	340/11	340/18	280/35	280/35	350/40	140/32	180/27	170/18	090/22	110/16	260/10	080/10
2012	110/18	350/23	310/26	250/45	230/30	330/55	160/26	270/22	120/26	030/16	360/10	360/08
2013	180/10	040/12	230/18	350/40	280/45	130/38	200/20	240/20	190/20	110/20	350/10	190/12

Table-1: Monthly maximum wind direction & speed.

61. **Rainfall**. Month-wise total amount of rainfall from January 1984 to December 2013 is shown in Table-2:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1984	51	1	Nil	019	304	821	292	321	167	54	Nil	Nil
1985	14	4	50	4	177	311	244	206	238	189	Nil	Nil
1986	9	Nil	1	90	136	337	338	177	637	183	159	6
1987	1	Nil	58	165	52	327	389	714	193	136	14	10
1988	Nil	27	93	148	298	518	257	312	117	87	100	Nil
1989	Nil	3	2	41	273	202	282	120	161	234	0	7
1990	Nil	5	189	19	163	342	392	106	202	94	126	10
1991	24	24	33	54	87	408	220	335	216	145	Nil	76
1992	38	66	Nil	20	204	303	317	147	203	36	3	Nil
1993	Nil	33	115	147	142	294	335	210	333	103	4	Nil
1994	12	29	1	51	175	371	220	255	89	64	2	Nil
1995	17	25	18	5	88	196	281	286	270	95	113	3
1996	21	20	22	90	146	350	289	362	195	257	5	Nil
1997	15	33	51	70	110	232	393	240	348	12	15	19
1998	44	26	141	130	222	158	184	209	168	88	118	Nil
1999	Nil	Nil	Nil	31	141	262	448	251	301	151	1	Nil
2000	2	19	7	71	172	500	283	279	413	145	Nil	Nil
2001	Nil	38	37	39	347	475	277	109	141	165	14	Nil
2002	18	Nil	22	134	267	539	519	258	224	32	112	Nil
2003	Nil	13	119	55	66	457	419	142	145	466	Nil	35
2004	3	Nil	10	72	176	303	398	400	856	259	Nil	Nil
2005	32	15	129	33	57	181	468	113	159	411	2	Nil
2006	Nil	Nil	1	52	302	231	479	318	366	7	4	Nil
2007	Nil	83	21	62	159	351	738	274	261	142	100	Nil
2008	70	38	38	38	213	268	415	200	415	199	Nil	Nil
2009	Nil	7	34	Nil	198	240	279	391	417	94	1	1
2010	Nil	22	11	50	274	297	168	211	186	116	4	32
2011	Nil	Nil	34	35	145	360	250	303	227	18	1	Nil
2012	31	4	5	92	26	253	304	185	271	71	61	2
2013	6	15	8	63	301	212	211	309	328	215	Nil	Nil

Table-2: Monthly total amount of rainfall (in mm)

62. **Rainy Days**. Monthly total numbers of rainy days from Jan 84 - Dec 13 are shown in Table-3:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1984	4	1	Nil	7	8	21	22	24	13	6	Nil	Nil
1985	4	2	4	2	15	18	20	23	16	3	Nil	Nil
1986	1	Nil	1	7	10	17	19	17	23	12	5	Nil
1987	1	Nil	7	10	6	11	24	17	13	4	3	1
1988	Nil	1	8	4	14	18	23	20	9	5	4	Nil
1989	Nil	2	2	3	9	15	18	16	7	9	Nil	2
1990	Nil	6	10	9	15	18	21	14	22	7	4	1
1991	3	3	2	6	7	20	20	22	15	10	Nil	3
1992	3	6	Nil	2	12	16	18	18	14	5	1	Nil
1993	Nil	2	4	6	16	15	19	24	22	11	1	Nil
1994	3	3	1	5	8	21	21	23	12	5	2	Nil
1995	4	3	1	2	7	17	23	20	14	7	6	1
1996	3	4	2	5	7	18	15	20	12	6	1	Nil
1997	2	3	4	8	10	21	24	24	18	1	3	3
1998	9	3	9	7	10	13	22	19	14	13	5	Nil
1999	Nil	Nil	Nil	4	12	18	19	18	16	15	1	Nil
2000	2	6	2	6	14	22	25	21	14	14	Nil	Nil
2001	Nil	2	6	7	15	25	26	17	13	11	2	Nil
2002	4	Nil	4	9	13	20	20	22	22	3	2	Nil
2003	Nil	2	9	6	9	22	21	19	21	17	Nil	4
2004	1	Nil	5	6	9	18	22	24	15	9	Nil	Nil
2005	3	2	6	3	6	14	25	12	18	15	1	Nil
2006	Nil	Nil	1	6	13	15	25	27	19	4	4	Nil
2007	Nil	6	6	6	11	19	25	20	22	9	4	Nil
2008	4	3	8	6	6	21	27	22	20	10	Nil	Nil
2009	Nil	1	2		11	10	22	24	12	17	1	1
2010	Nil	4	1	4	11	15	18	16	16	8	3	3
2011	Nil	Nil	7	5	10	18	17	20	19	2	1	Nil
2012	5	1	1	9	6	15	23	18	16	8	5	1
2013	1	2	4	10	21	18	25	26	9	19	Nil	1

Table-3: Monthly total no of rainy days

CHAPTER – VI : CONCLUSION AND RECOMANDATIONS

CONCLUSION

1. Weather forecasting has enormous importance in all areas of military operations as well as their supporting personnel and equipments. The accuracy of weather forecast of a particular area depends largely upon the knowledge of the forecaster about the prevailing weather conditions and the climatology of that area. For providing reliable climatology of different weather elements, 30 years real time data have been analysed for obtaining average conditions of different weather elements like temp, fog, thunderstorm/Nor'wester, heat wave, cold wave, rainy days, break monsoon etc. These weather elements are indispensable for planning of flying effort, air operations, ground operations, ceremonial planning etc and their executions throughout the year. The extreme conditions of different weather elements found within last 30 years have been shown in the table. Here, the temp data have been studied in the chapter "Temperature Psychotherapy" and the data of max, min temp for monthly and fortnightly have been used for obtaining the average conditions. The fog data have been processed in the chapter "Fog Investigation" for obtaining the monthly foggy days and the average cessation time of fog in different months. The occurrence of thunderstorms/Nor'wester have been analysed in the chapter "Thunderstorm Behavior" for obtaining the frequency, time of occurrence, susceptible dates of occurrence in different months and findings are appended in the "Climate of Jessore at a glance". The amount of rainfall and rainy days data have been analysed in the chapter "Precipitation Study" for obtaining yearly, monthly, pentad and daily average; to obtain susceptible days of continuous rain; break in monsoon etc. In the study it was found that the yearly amount of rainfall and no rainy days are highly variable. However, the yearly average amount of rainfall was found **1703mm** and the average no of rainy days was **111.1 days**. The findings of the study have been shown below as "climate of Jessore at a glance".

Climate of Jessore at a Glance

2. After analyzing all the data, the findings have been shown graphically and the inferences are also discussed below the graphs of each element for easy understanding. However, the findings are appended below in brief:

a. **Temperature, Heat Wave and Cold Wave Conditions**. The temperature data study reveals that the hottest month of the year is April and coldest month is January. The significant findings regarding temp are as fol:

(1) April is the hottest month in the year with monthly mean max temp 35.7ºc in April and the 2nd fortnight of April is the hottest period in the year with fortnightly mean max 36.2ºc. The highest max temp within last 30 years was found 43.2ºc on 09 May 09.

(2) The mean max temp starts rising from 16 January to 31 March with a rate of 2ºc - 3ºc per fortnight; thereafter it become 1ºc – 2ºc per fortnight till 30 April (2nd fortnight of April).

(3) The temp ≥38ºc was found average 06 days in April (max 20 days in 1992) and 04 days in May (Max 14 days in 2004). This temp also found in March (0.8 day) and June (1.4 days). Therefore, most of the severe (40-42ºc) heat waves were found in the 2nd fortnight of April (11 heat wave), then the 1st fortnight of May (05 heat wave); severe heat wave also found in the 1st fortnight of April (02 heat wave), 2nd fortnight of May (04 heat wave) and 1st fortnight of June (01 heat wave).

(4) During heat wave conditions, flying becomes restricted, people become sick due to heat stroke and human performance reduces due to sweating.

(5) January is the coldest month in the year with monthly mean min temp 10.7ºc and the 1st fortnight of January is the coldest fortnight with fortnightly mean minimum 10.4ºc. The lowest min temp within last 30 years was found 4.2ºc on 09 Jan 13.

(6) The mean minimum temp starts falling from 01 October and continues till 16 January with a rate of 2ºc – 3ºc per fortnight; thereafter it increases in the same rate till 31 Mar.

(7) The temp ≤10ºc was found average 04 days in December (max 13 days in 2007 & 2010), 12 days in January (max 23 days in 2003) and 01 day in February (max 08 days in 2008). The maximum no of moderate (6-8ºc) to severe (4-6ºc) cold wave was found in 1st fortnight of January (12 cold wave), then in 2nd fortnight (09 cold wave); cold wave also found in 2nd fortnight of December (04 cold wave).

(8) A cold wave can cause death and injury to livestock and wildlife. Exposure to cold mandates greater caloric intake for all animals, including humans. The belief that more deaths are caused by cold weather in comparison to hot weather is true as a result of cold, flu, pneumonia, etc.

b. **<u>Persistence of Fog in Winter</u>**. Fog starts in September and continues till last of April. The average no of foggy days from September to April are 01, 03, 09, 18, 20, 14, 08 and 02 days respectively. Monthly average end time of fog is being calculated from November to March. Because, the foggy days in other months (Apr, Sep and Oct) are very less in number and short in duration. The persistence of fog in different months are as fol:

(1) In November the average foggy days was 09 (Fig-11). Among these, average 05 days (53%) fog persists up to 0800F and 04 days (47%) up to 0900F.

(2) In December the average foggy days was 18 (Fig-11). Among these days, average 12 days (65%) fog persists up to 1000F, 5.75 days (32%) up to 0900F and only 0.25 days (3%) up to 0800F.

(3) In January the average foggy days was 20 (Fig-11). Among these days, average 08 days (40%) fog persists up to 0900F, 10 days (50%) up to 1000F and 02 days (13%) up to 1100F.

(4) In February the average foggy days was 14 (Fig-11). Among these days, average 11.5 days (82%) fog persists up to 0900F and 2.5 days (18%) up to 1000F.

(5) In March the average foggy days was 08 (Fig-11). Among these, average 1.3 days (17%) fog persists up to 0800F, 06 days (73%) up to 0900F and only 0.7 days (10%) up to 1000F.

(6) The activities which are related with visibility like flying, driving etc needs to plan according to the cessation time of fog in different months. Flying should be planned after 1000F in December and January. In the same way, flying should be planned after 0900F in February and March.

c. **Occurrence of Nor'wester**. The thunderstorms occur in pre-monsoon season (February to May), most of them are associated with Nor'wester but some of them may not be Nor'wester. In this study, Norw'ester has been considered when wind speed was 22 kts or more. The month-wise susceptible dates of Nor'wester are as fol:

(1) In February, the average frequency of Nor'wester is 0.5 (Fig-18). The susceptible days for occurrence of Nor'wester are 19th, 20th, 25th and 28th February (Probability 10% -15%).

(2) In March, the average frequency of Nor'wester is 2.2 (Fig-18). Nor'westers mainly starts from 22 March, though few may occur early also. The susceptible dates for occurrence of Nor'wester are 23rd, 24th, 27th, 29th and 30th March (probability 20% - 30%).

(3) In April, the average frequency of Nor'wester is 5.5 (Fig-18). The susceptible days of nor'wester can be considered as 22nd, 24th, 26th, 28th, 29th and 30th April and the probability is around 30%.
.

(4) In the month of May, the frequency of Nor'wester is highest (average 8.5). The susceptible days of nor'wester can be considered as 3rd, 5th, 10th, 13th, 15th, 17th, 24th, 25th, 26th and 27th April (probability 30% - 40%).

(5) Nor'wester occurs in the 1st and 2nd week of June also, till monsoon is set over the country. The 4th, 5th, 7th, 10th and 11th June are vulnerable for occurrence of Nor'wester

(6) The maximum no of Nor'wester (254 out of 506 Nor'wester) occurred within the time 1200Z to 1800Z (50%) and 25% Nor'wester (129 out of 506 Nor'wester) occurred within 0600Z - 1200Z. (Fig-19)

(7) The susceptible days for nor'wester need to be more cautious for any type of outdoor activities including flying, driving or any public gathering/ceremony, particularly within 1200Z – 1800Z.

d. **Break in Monsoon**. Monsoon break is the break of rainfall during monsoon period. After onset of monsoon, the susceptible dates of break in different months are as fol:

(1) The onset and withdrawal date of monsoon is found 12th June and 10th October respectively over Jessore area.

(2) In July, 08 - 09 Jul are susceptible for one break and 16 - 17 Jul are susceptible for another break in monsoon.

(3) In August, 06–07 Aug, 16–18 Aug and 23–24 Aug are susceptible for break in monsoon.

(4) In September, 01–02 Sep, 06–07 Sep, 17–21 Sep and 28– 30 Sep are susceptible for break monsoon.

(5) During the break monsoon periods, rainfall ceases and cloud decreases. As a result, these periods could be utilized for compensating the loss occurred in the days when rainfall was continuous.

e. **Continuous Rainy days in Monsoon**. During monsoon, there are few days when it rains cats and dogs and all activities are disrupted. The vulnerable days in different months are as fol:

(1) After onset of monsoon, the 17, 20, 22, 25 and 29th June are susceptible for continuous rainy days, when normal activities may be hampered due to rain.

(2) In July, the 04, 15, 19, 21, 23, 28 and 29 July are susceptible for continuous rainy days.

(3) In August, the 02, 14, 15, 19 and 28 August are susceptible for continuous rainy days.

(4) In September, the 05, 09 and 11 September are susceptible for continuous rainy days.

(5) The continuous rainy days affect the flying, ceremonial activities and public life in a greater extent. During these days all outdoor activities bring to a halt including flying.

Recommendations

1. To ensure the effective use of climatological records for Air force operations, Military and Marine operations, the study report recommends the followings:

a. The result of the study may be taken into cognizance during the planning of any flying effort, air operation, ground operation and ceremonial activities around Jessore area.

b. Necessary steps may be taken for preservation of long term Met data centrally for all bases/unites of BAF including isolated Radar unites, so that it can be compiled for climatology of whole Bangladesh.

c. Similar study may be encouraged for all bases with available data for finding out the local climate.

d. Steps may be taken to disseminate sister services for utilization in their operations and for the greater benefit of the country.

e. Steps may be taken for acquisition of Met data using installations of sister services from remote areas where no BAF installations are available by providing training to them.

f. Steps may be taken to disseminate BMD for updating in media.

Jessore

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Wing Commander

November 2014

OIC Met Sqn, BAF MTR

(Total WordsApproximately-12479)

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GPCP Pentad Precipitation Analyses: An Experimental Dataset Based on Gauge Observations and Satellite Estimates
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ONSET AND END OF THE FIRST RAINY SEASON IN SOUTH CHINA
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Extended range prediction of active-break spells of Indian summer monsoon rainfall using an ensemble prediction system in NCEP Climate Forecast System
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